CS1675/2075: Introduction to Machine Learning

Week 7: Linear Models Deep Dive

Fall 2024

So far this semester we have covered:

- A general overview of machine learning -> RESAMPLING and OVERFITTING!

Applied Machine Learning

Distribution fitting

Supervised learning deep dive

Unsupervised learning

So far this semester we have covered:

- A general overview of machine learning
- Learning unknown constants from data -> UNCERTAINTY and OPTIMIZATION!

Applied Machine Learning

Distribution fitting

Supervised learning deep dive

Unsupervised learning

Today, we start discussing predictive modeling in depth.

- The goal is to learn relationships or trends between a response and inputs

Applied Machine Learning

Distribution fitting

Supervised learning deep dive

Unsupervised learning

Specifically, we will begin with **LINEAR MODELS**

Applied Machine Learning

Distribution fitting

Supervised learning deep dive

Unsupervised learning

Linear models
Generalized Linear models
Non-linear methods

A linear model to predict a response, y, based on a single input, x

- Consider that we have N observed input-output pairs:

$$\{x_n, y_n\} \text{ for } n = 1, \ldots, N$$

- The response is continuous and for now we will work with a continuous input.

Is this a regression or classification problem?

A linear model to predict a response, y, based on a single input, x

- Consider that we have N observed input-output pairs:

$$\{x_n, y_n\} \text{ for } n = 1, \dots, N$$

- The response is continuous and for now we will work with a continuous input.

The response is continuous. Linear models are applied to regression problems.

Start simple: Use a linear relationship between the response and the input. We wrote this model earlier in the semester as:

$$y_n = \beta_0 + \beta_1 x_n + \text{error}$$

The error term is typically written as: ϵ_n

Let's consider a linear <u>relationship</u> between the response and the input. Use "epsilon" instead of "error":

$$y_n = \beta_0 + \beta_1 x_n + \epsilon_n$$

The errors are normally distributed:

$$\epsilon_n \sim \text{normal}(0, \sigma)$$

However, I do NOT like this notation!

$$y_n \cancel{=} \beta_0 + \beta_1 x_n + \epsilon_n$$

We need to accept the fact that models are APPROXIMATIONS!

The errors are normally distributed:

$$\epsilon_n \sim \mathrm{normal}(0, \sigma)$$

What does it mean that errors are normally distributed? How do the observations relate to the model?

This formulation allows us to break up the model into TWO distinct parts

- The complete model consists of a **DETERMINISTIC** trend and a **STOCHASTIC** relationship

$$y_n \mid x_n, \beta_0, \beta_1, \sigma \sim \text{normal}(y_n \mid \underbrace{\beta_0 + \beta_1 x_n}, \sigma)$$

The **mean** trend of the response as a function of the input!

If we know β_0, β_1 and x_n we will always get the same value for the **mean**!

This formulation allows us to break up the model into TWO distinct parts

- The complete model consists of a **DETERMINISTIC** trend and a **STOCHASTIC** relationship

$$y_n \mid x_n, \beta_0, \beta_1, \sigma \sim \mathrm{normal}(y_n \mid \mu_n, \sigma)$$

$$\mu_n = \beta_0 + \beta_1 x_n$$

The mean trend μ_n will also be referred to as the **linear predictor** for the linear model

This formulation allows us to break up the model into TWO distinct parts

- The complete model consists of a **DETERMINISTIC** trend and a **STOCHASTIC** relationship

$$y_n \mid x_n, \beta_0, \beta_1, \sigma \sim \boxed{\text{normal}}(y_n \mid \boxed{\mu_n}, \boxed{\sigma})$$

$$\mu_n = \beta_0 + \beta_1 x_n$$

The **STOCHASTIC** portion describes how the response varies around the mean trend!

The response is **NORMALLY** distributed around a mean.
The mean is a function of the input, thus the mean changes as the input changes.
The variability around the mean is controlled by σ! Higher σ, higher **noise**!

This formulation allows us to break up the model into TWO distinct parts

- The complete model consists of a **DETERMINISTIC** trend and a **STOCHASTIC** relationship

$$y_n \mid x_n, \beta_0, \beta_1, \sigma \sim \boxed{\text{normal}}(y_n \mid \boxed{\mu_n}, \boxed{\sigma})$$

$$\mu_n = \beta_0 + \beta_1 x_n$$

Does the variability or noise around the mean trend change with respect to x?

I also prefer writing the model with this notation because it is the <u>LIKELIHOOD</u>

- The likelihood of the n-th observation's response given the input x_n, the parameters $\boldsymbol{\beta} = \{\beta_0, \beta_1\}$ and noise σ.

$$\boxed{\begin{gathered} y_n \mid x_n, \beta_0, \beta_1, \sigma \sim \text{normal}(y_n \mid \mu_n, \sigma) \\ \mu_n = \beta_0 + \beta_1 x_n \end{gathered}}$$

Remember there are N observations

- We have focused on an individual observation's likelihood.

- We assumed the observations are **conditionally independent** **given** **the input and parameters**.

- Allows factoring the joint distribution between all observations into the product of N individual likelihoods.

Remember there are N observations

- Using vector notation, the vector of responses and inputs:
- $\mathbf{y} = [y_1, y_2, \ldots, y_n, \ldots y_N]^T$ and $\mathbf{x} = [x_1, x_2, \ldots, x_n, \ldots, x_N]^T$

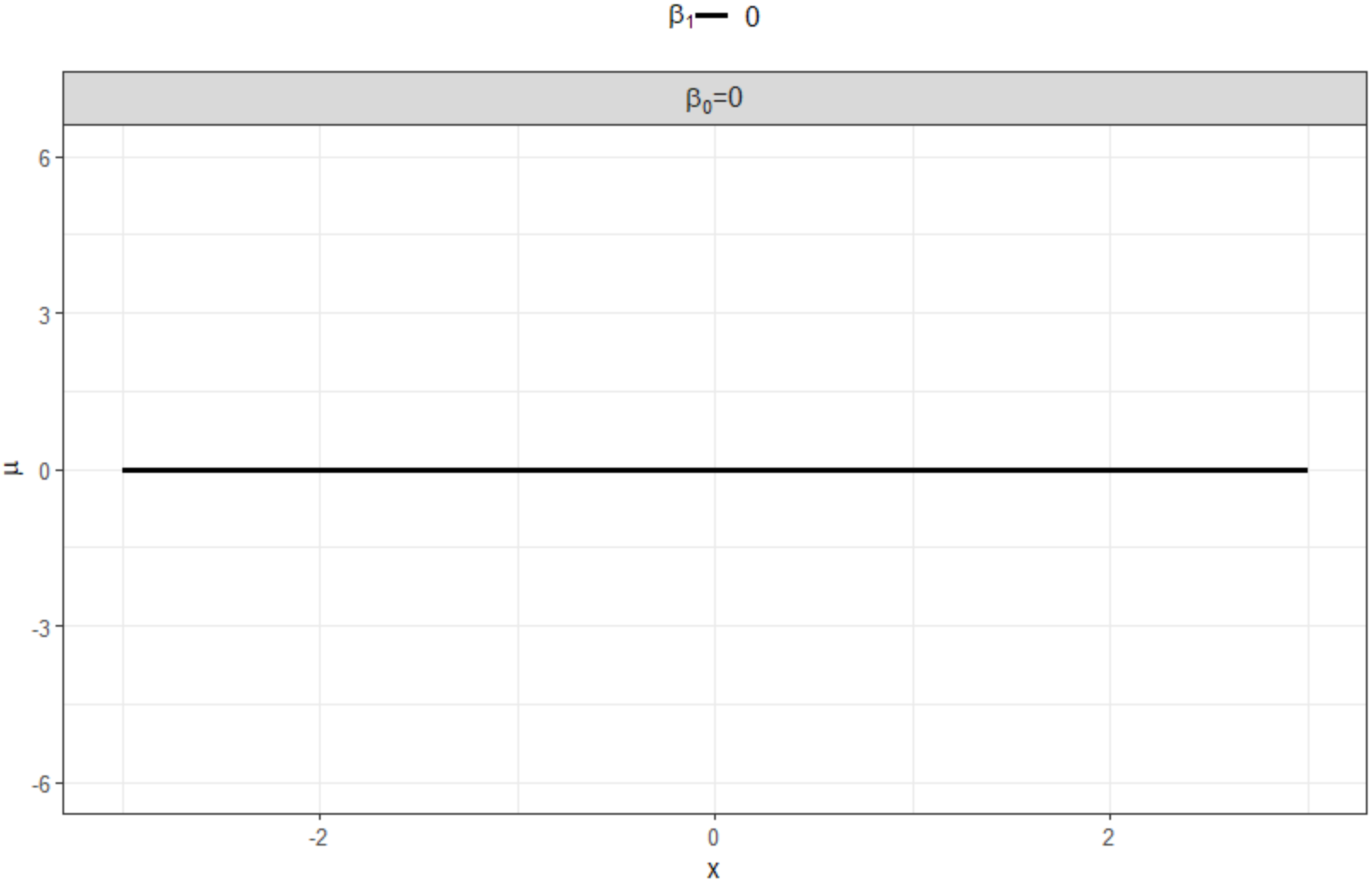
$$p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma) = \prod_{n=1}^{N} \{p(y_n \mid \mu_n, \sigma)\} = \prod_{n=1}^{N} \{\mathrm{normal}(y_n \mid \mu_n, \sigma)\}$$

$$\mu_n = \beta_0 + \beta_1 x_n$$

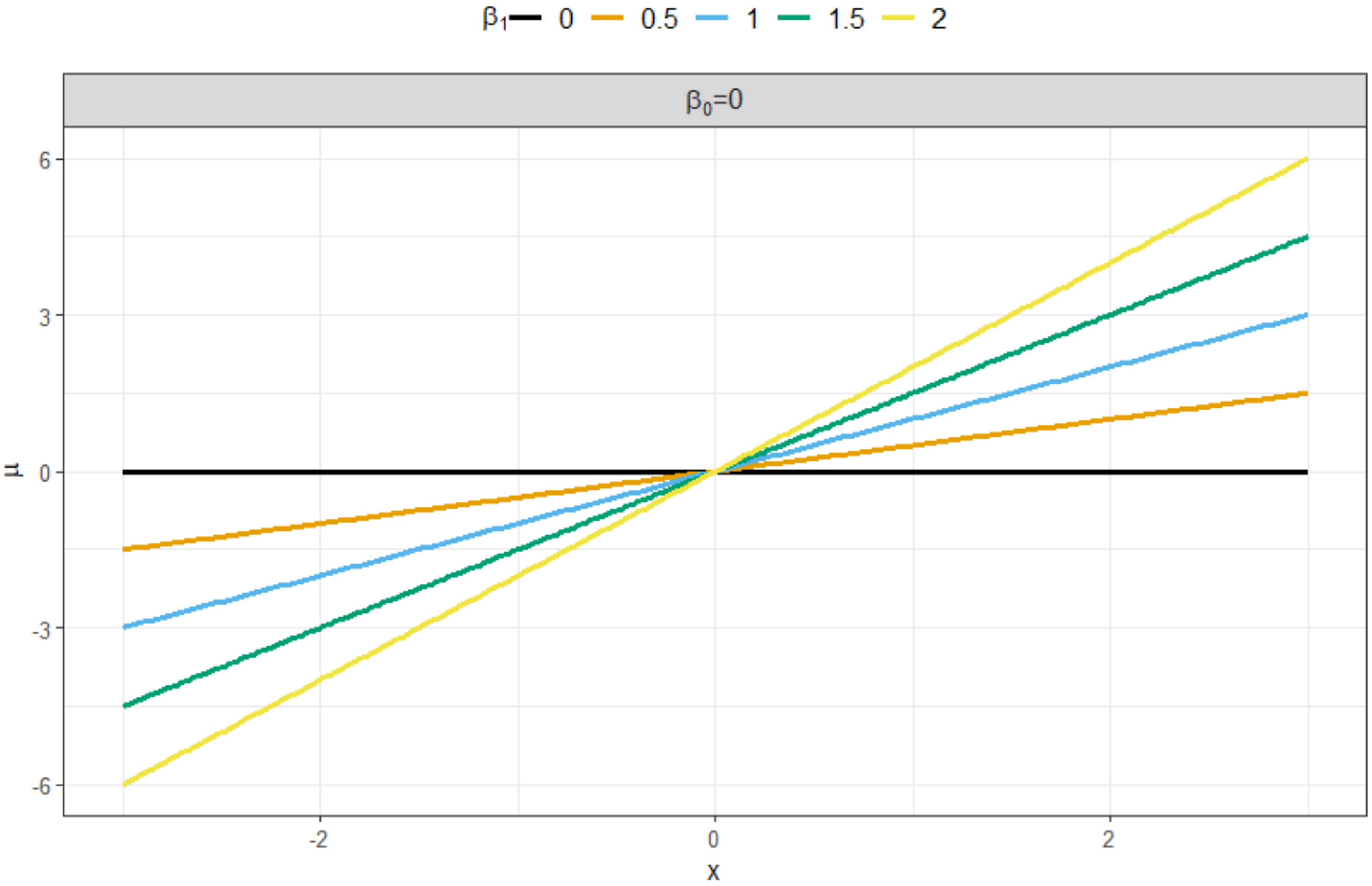
Trends: start with the mean, μ, wrt x

- β_0 is the intercept and β_1 is the slope.
- Consider an input, x, between -2 and +2.
- What happens if $\beta_0 = 0$ and $\beta_1 = 0$?

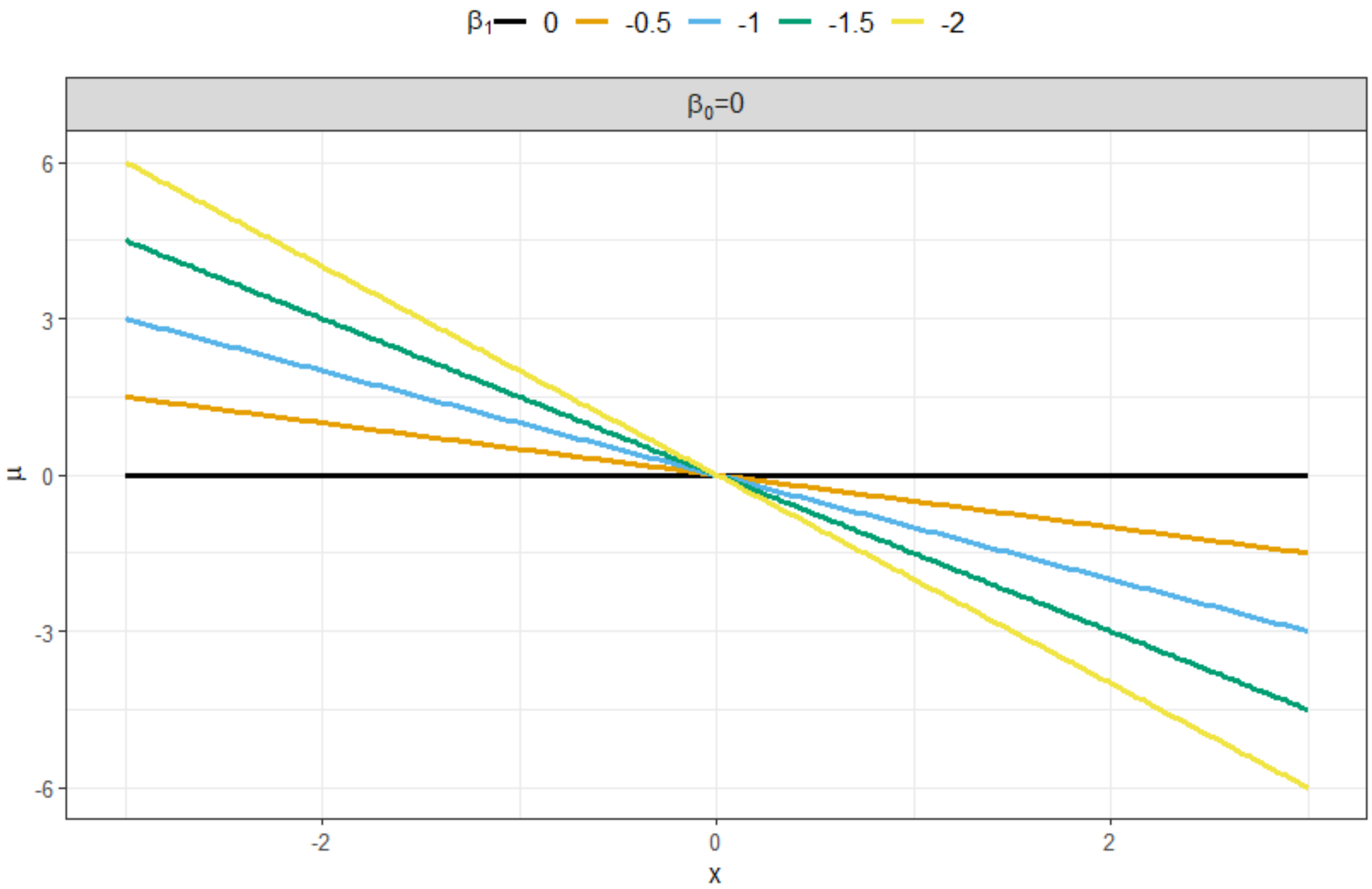
There is no trend between the mean and the input!



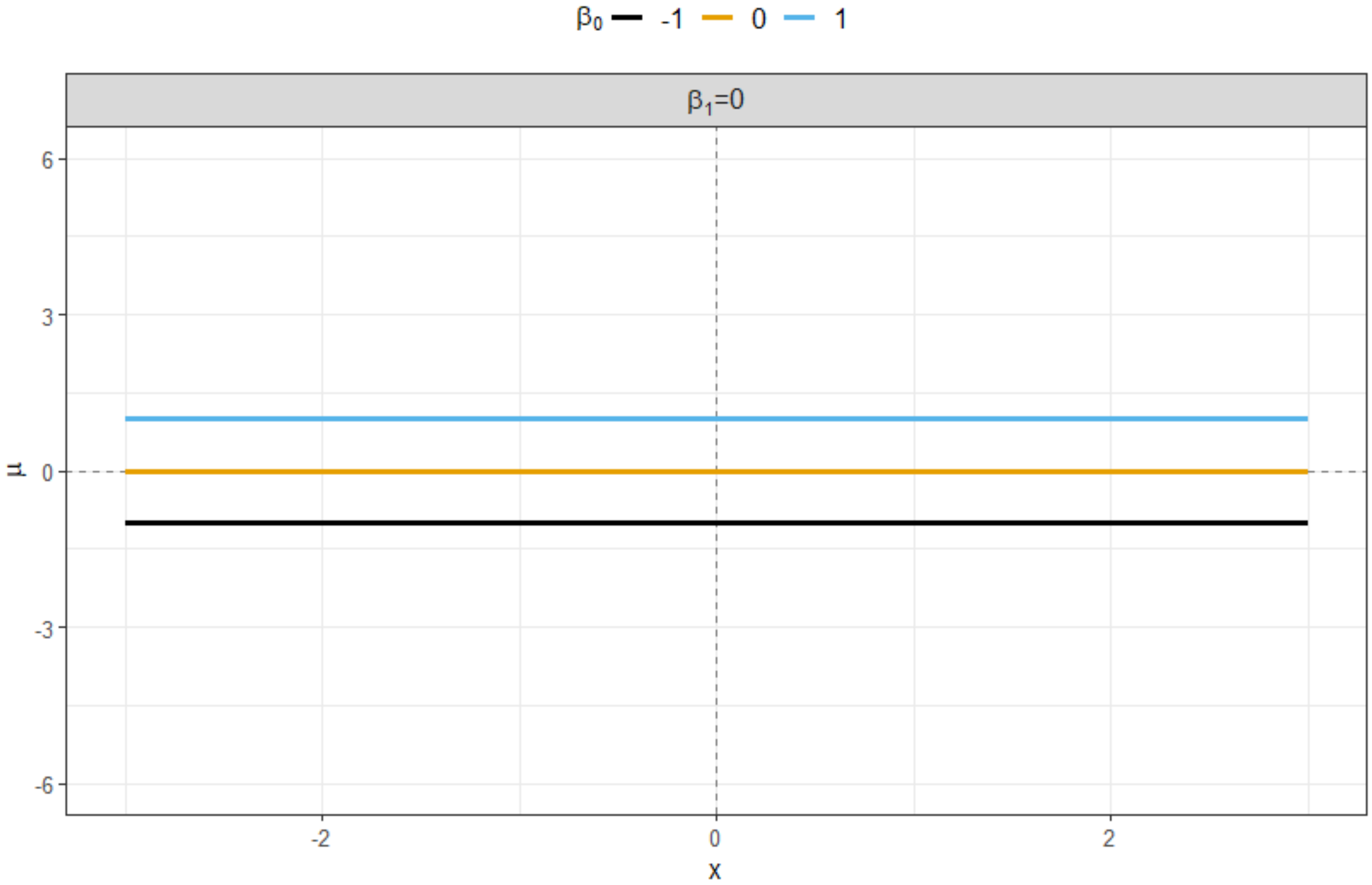
As the slope increases…



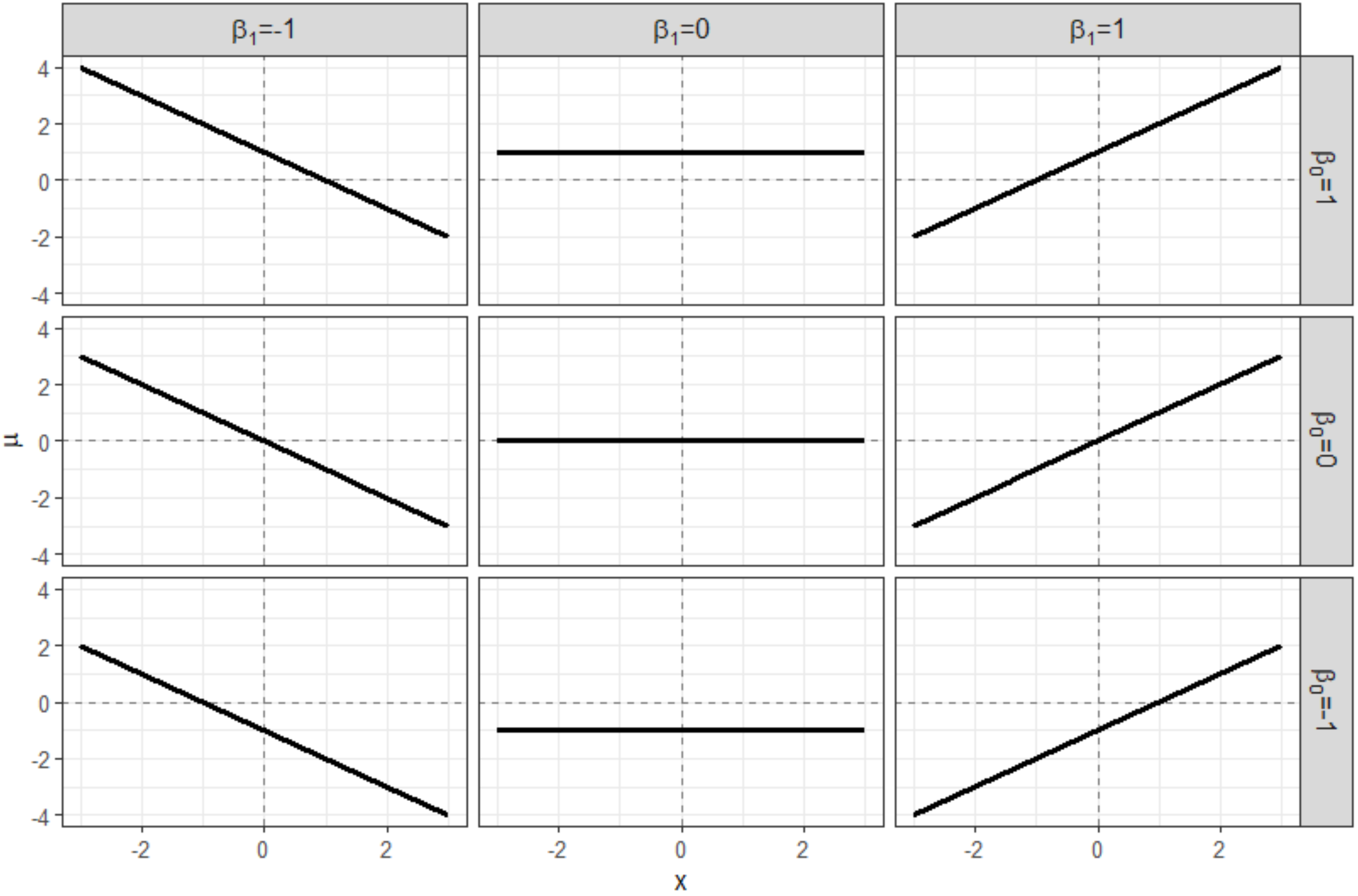
Negative slopes



Changing the intercept…



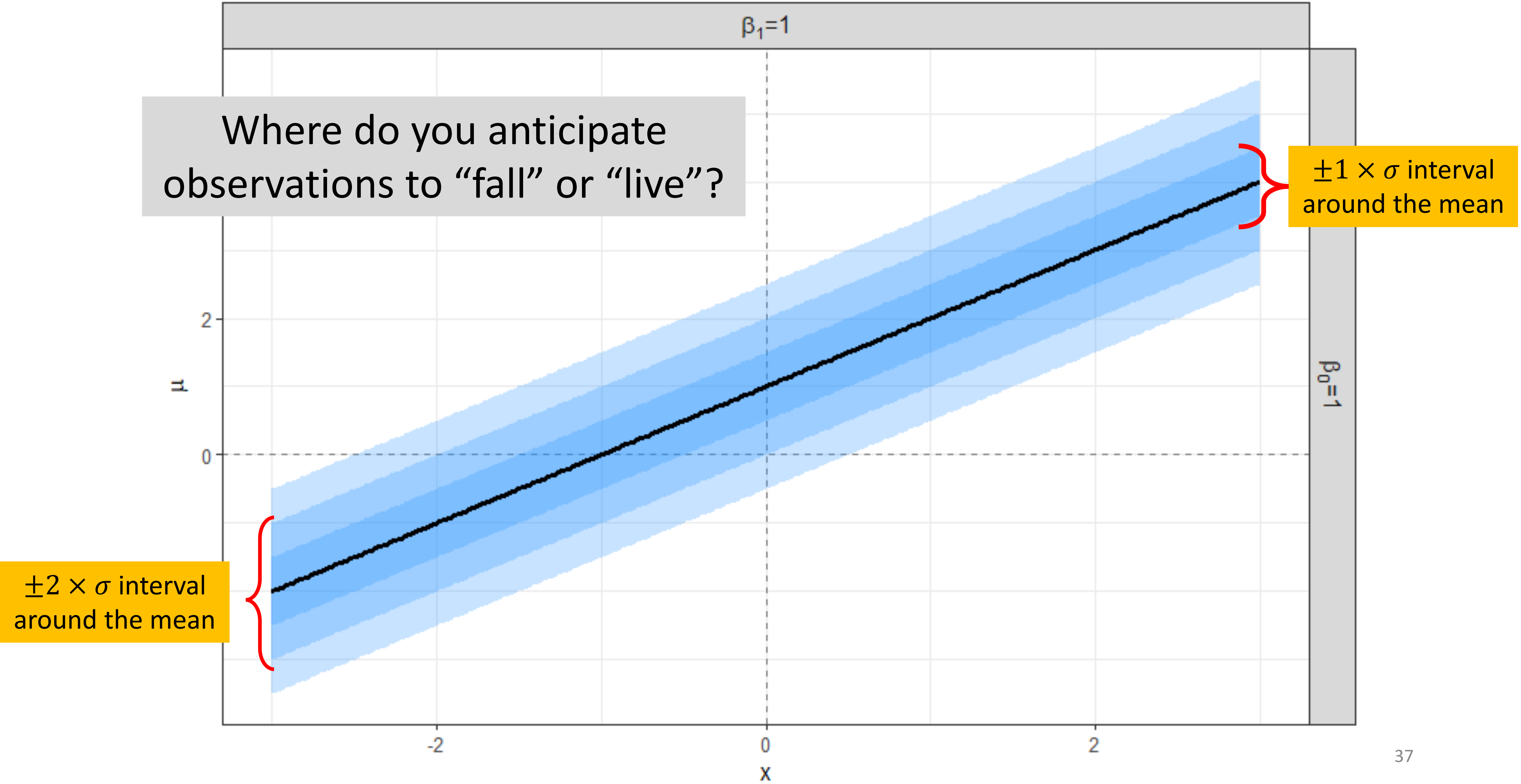
Changing the intercept and slope together



Observations are normally distributed around the mean with standard deviation σ.

- We will assume that $\sigma = 0.5$ for visualization purposes.

- The variability or **uncertainty** will be represented by overlapping ribbons to capture the $\pm 1\sigma$, $\pm 2\sigma$, and $\pm\, 3\sigma$ intervals.



β_1=1
β_0=1
Where do you anticipate observations to "fall" or "live"?
$\pm 1 \times \sigma$ interval around the mean
$\pm 2 \times \sigma$ interval around the mean
2
0
-2
0
2
μ
x

Let's generate random draws!

- Specify β_0 and β_1.
    - Specifically we will set both equal to 1.

```
b0 <- 1
b1 <- 1
```

- Specify σ.
    - Specifically we sill set equal to 0.5.

```
sigma <- 0.5
```

- Specify $\mathbf{x} = [x_1, \dots, x_N]^T$.
    - We will use 9 evenly spaced points between -2 and 2.

```
x <- seq(-2, 2, by = 0.5)
```

Let's generate random draws!

- Calculate the mean, μ_n, associated with each x_n GIVEN $\boldsymbol{\beta}$.

$$\mu_n = \beta_0 + \beta_1 x_n$$

```
mu <- b0 + b1 * x
```

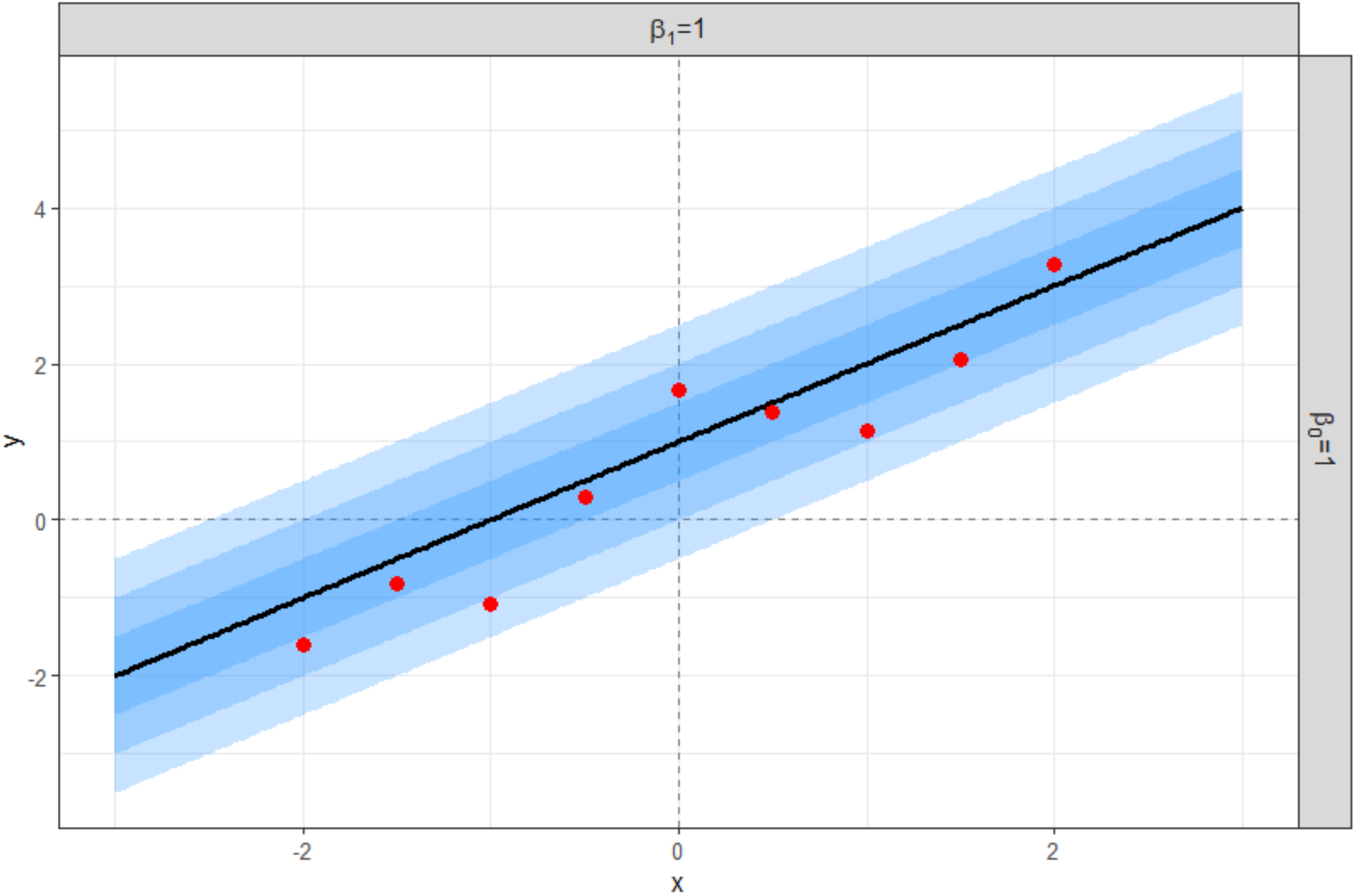
- Generate normal random numbers:

$$y_n \mid \mu_n, \sigma \sim \mathrm{normal}(y_n \mid \mu_n, \sigma)$$

```
y <- rnorm(n = length(x), mean = mu, sd = sigma)
```

The mean DEPENDS on the INPUT!!!!

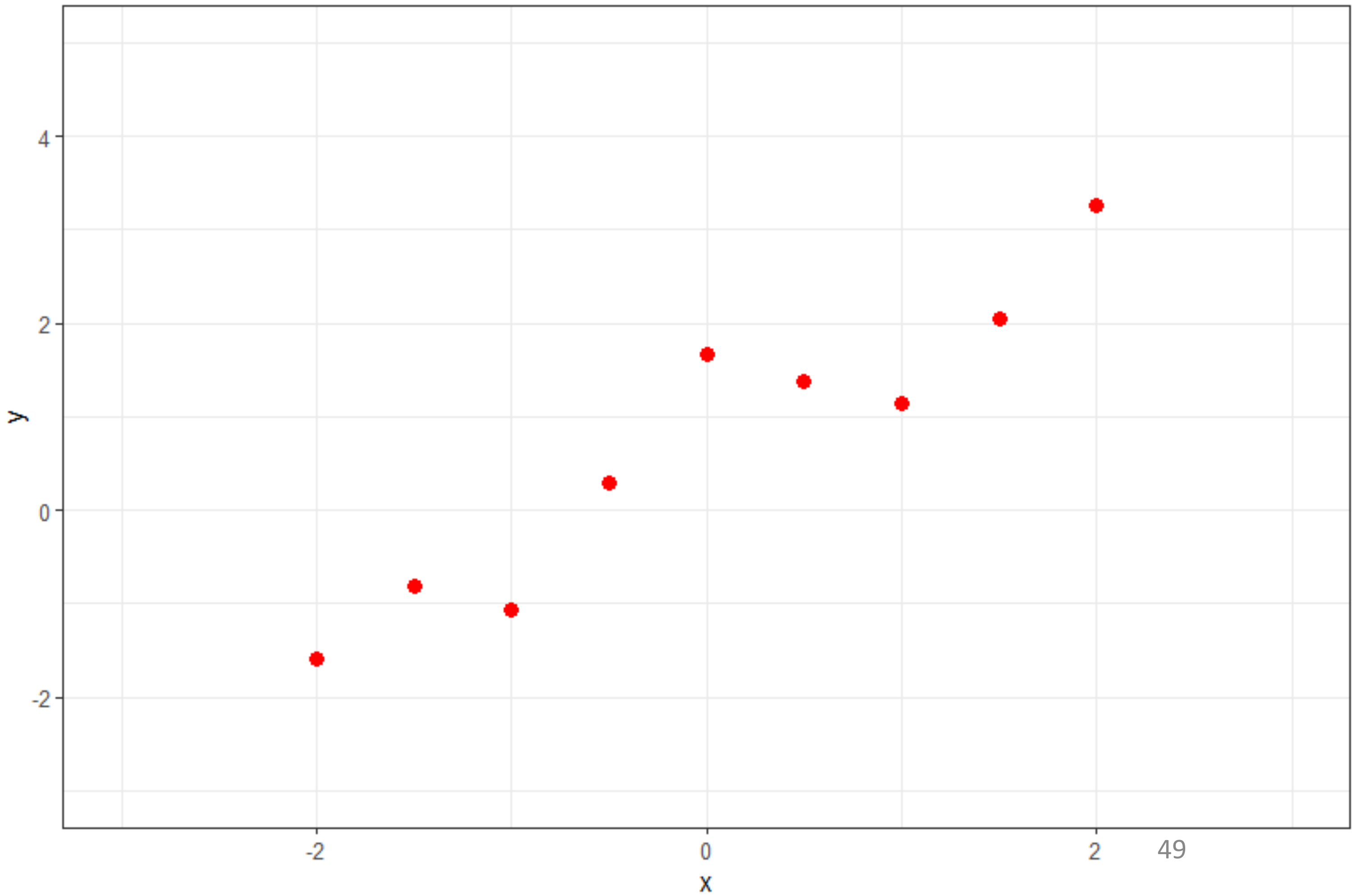
Plot the random observations as red markers on top of the mean trend and ribbons



We know the observations are generated from a linear relationship between the response and the input…

However, if we only had random observations…

Would we be tempted to think the relationship is non-linear…

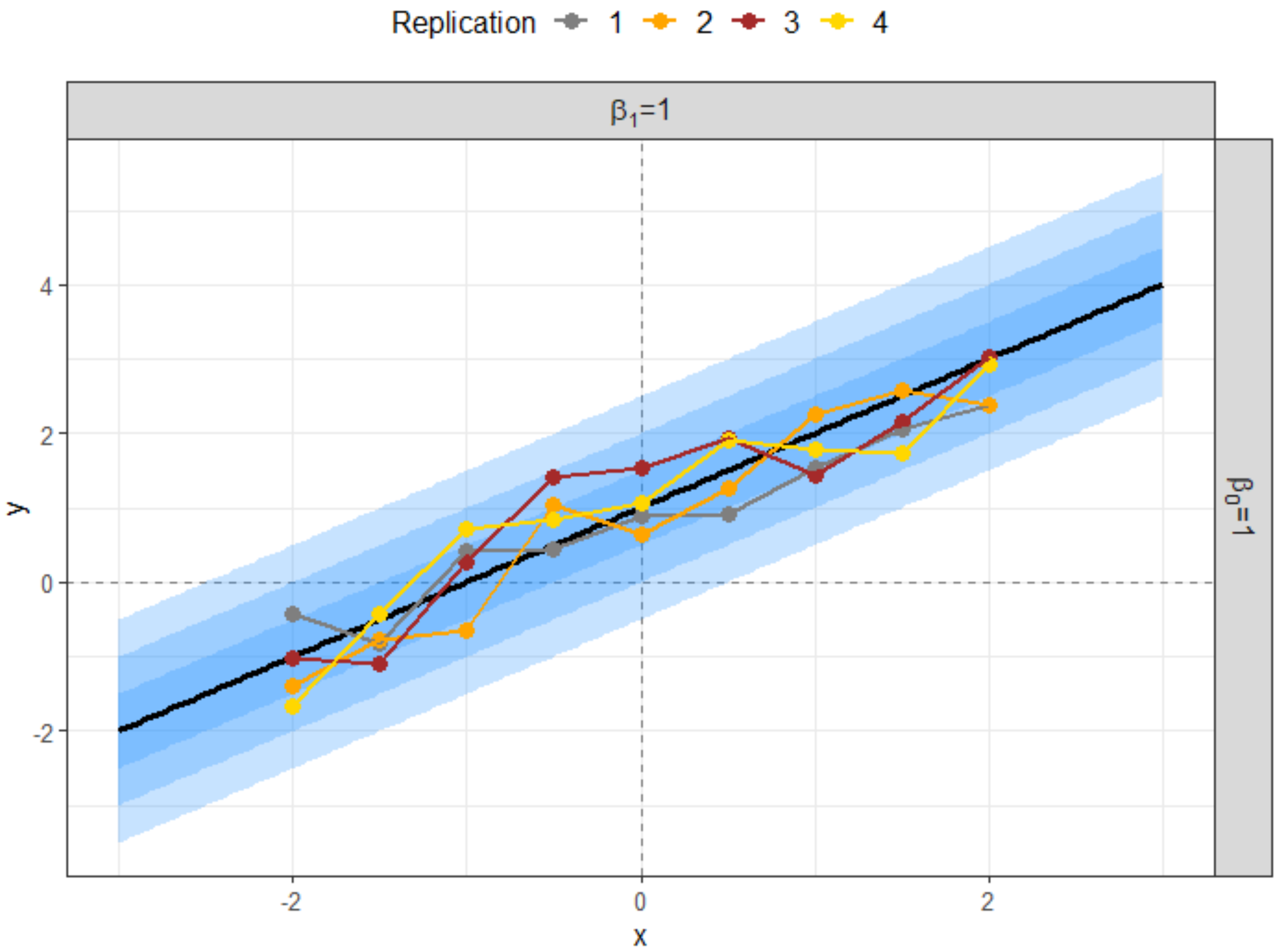


What happens if we increase the sample size at each input location?

- Rather than generating one random value per input location, generate 4 repeats or **replications** per x value.

- The next slide connects the observations associated with each replication just to help the visualization.

- Remember, we generated random draws based on a **deterministic linear relationship** between the mean response and input!



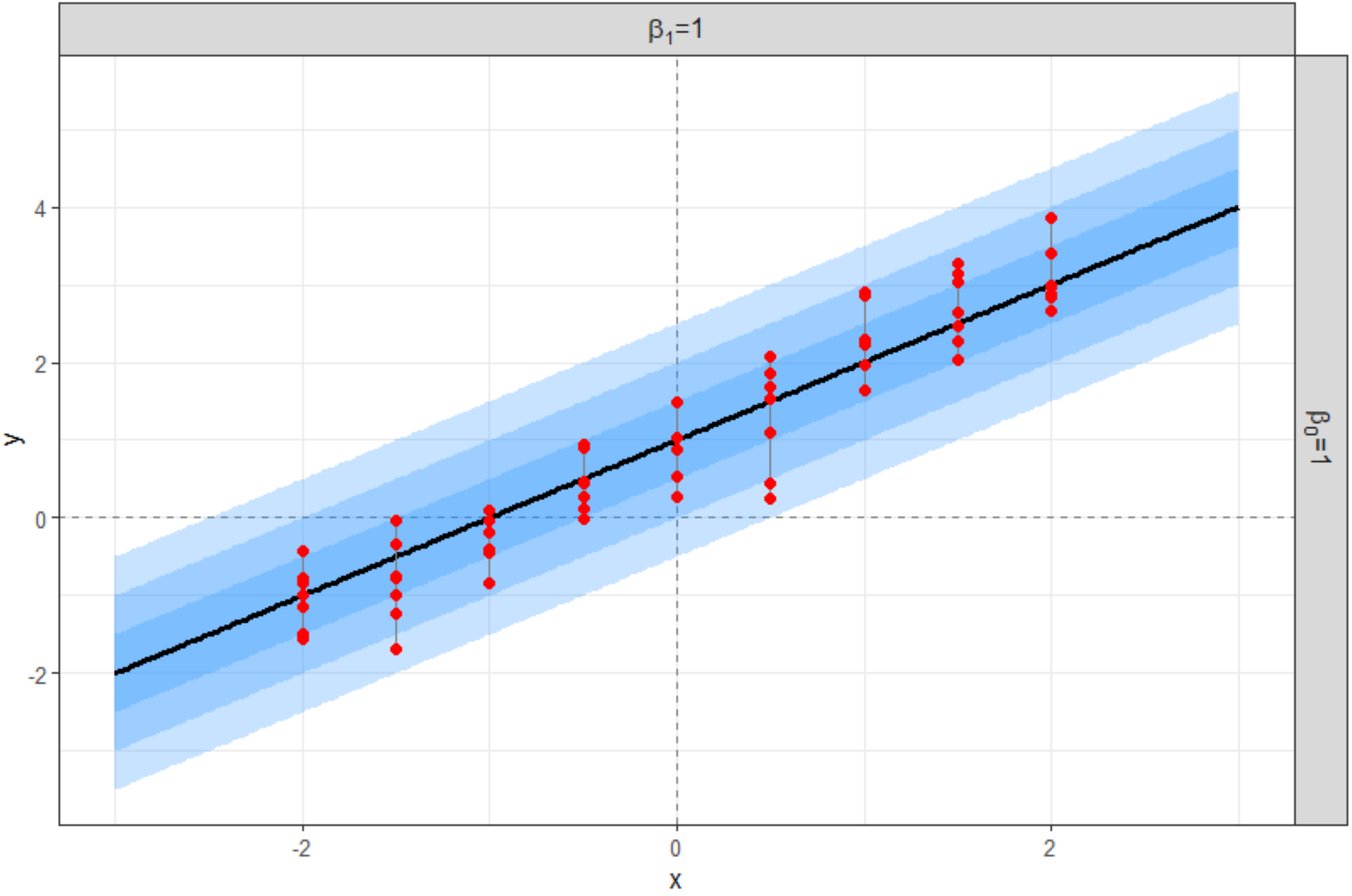
Replication 1 2 3 4
β_1=1
β_0=1
4
2
0
-2
y
-2
0
2
x

Increase sample size from 4 to 7

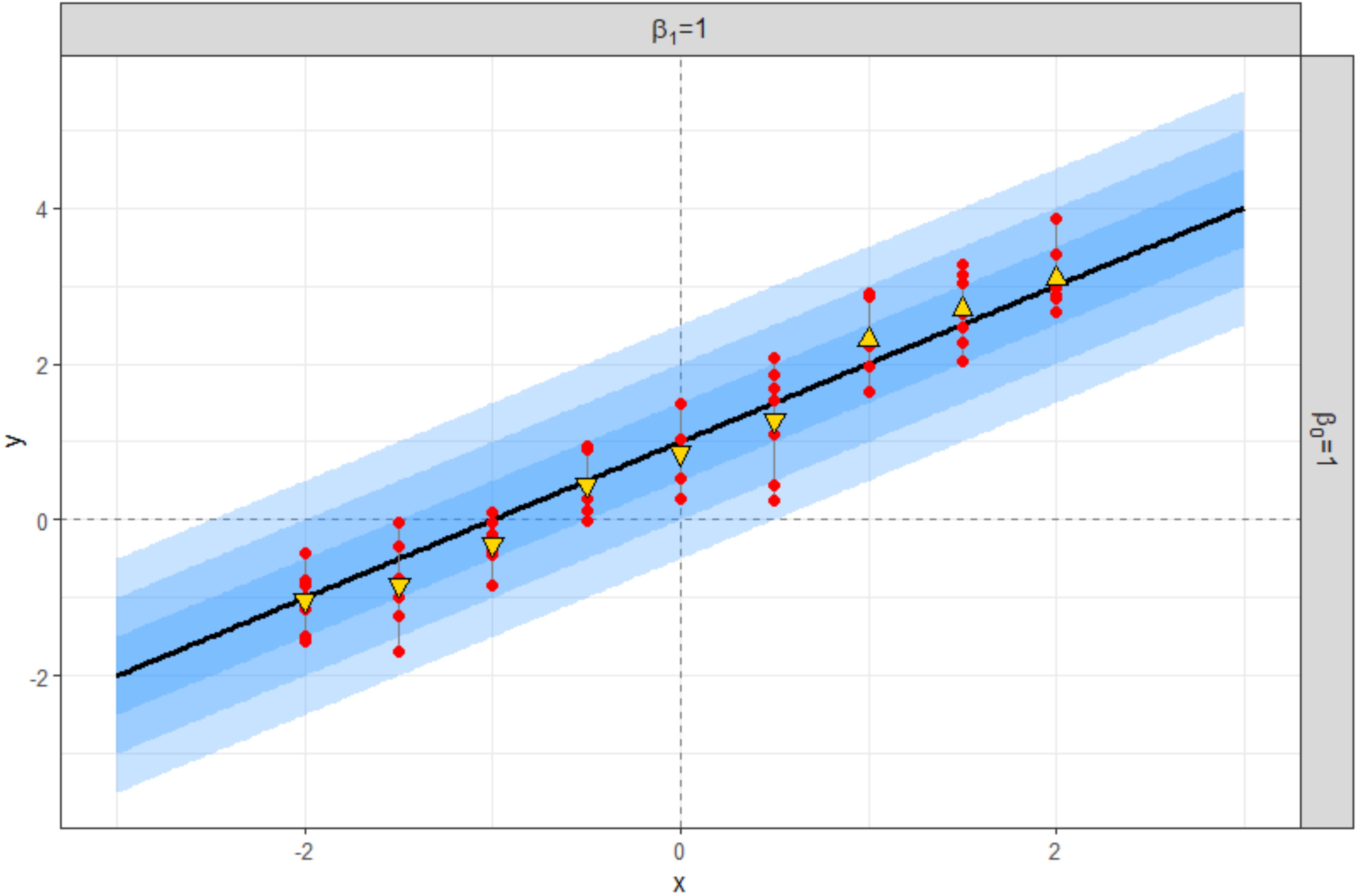
- Instead of connecting markers for each replication (connect across the input x), connect the min and max observed responses at each input location.

- The min and max display the empirical range at each input location.

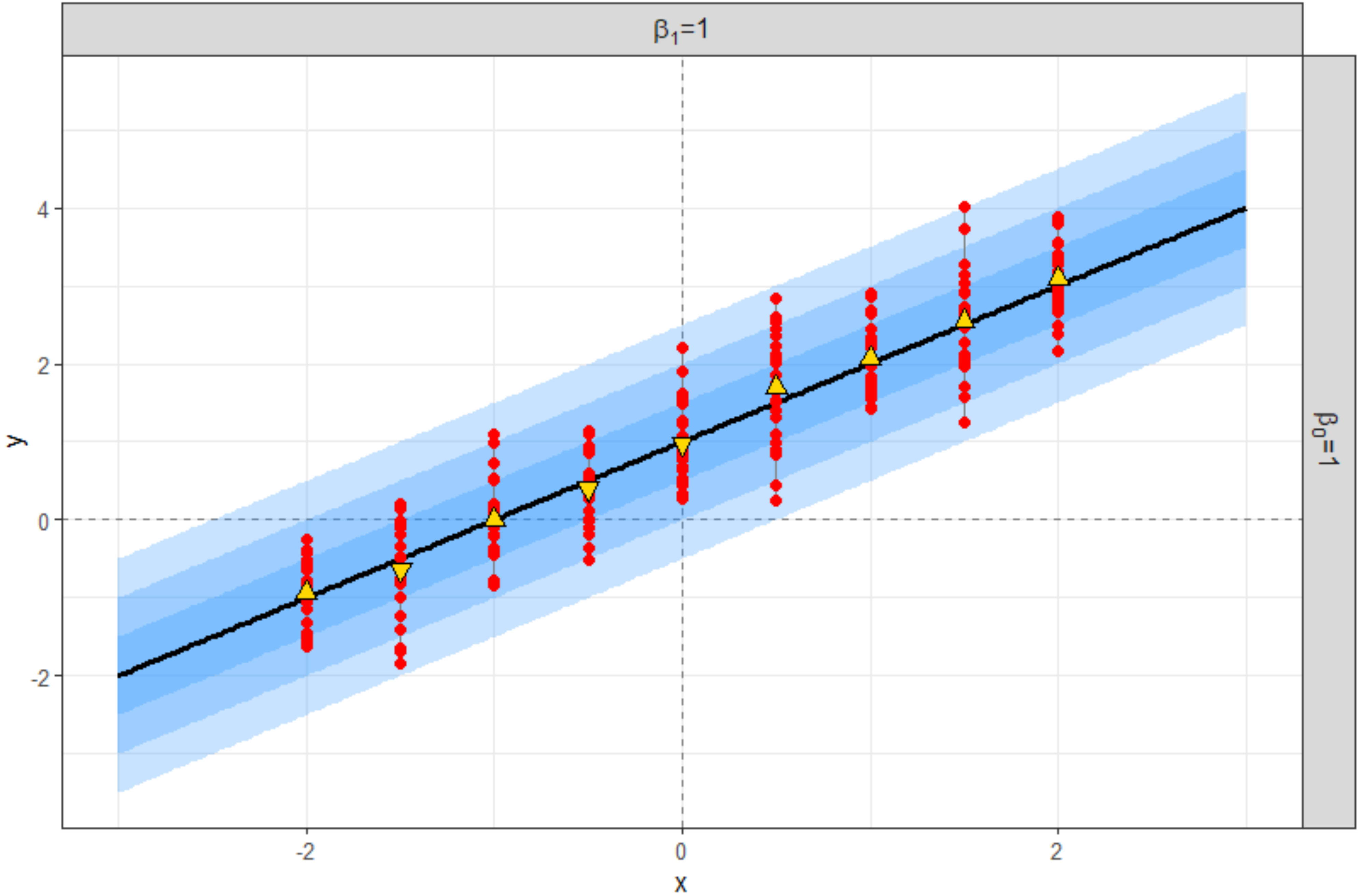
Increase the sample size from 4 to 7 replications per input location



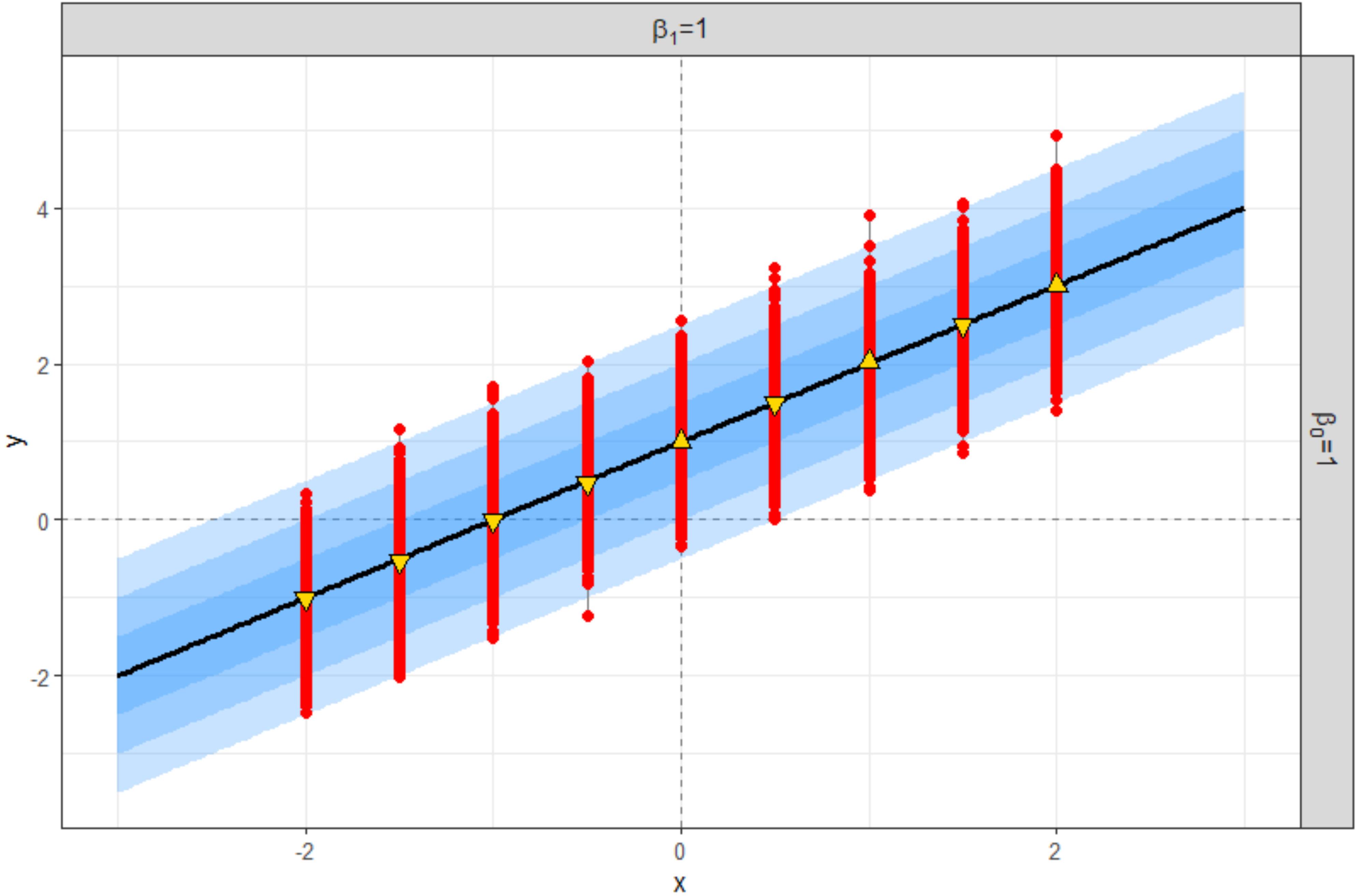
Include the response **sample average** at each input location



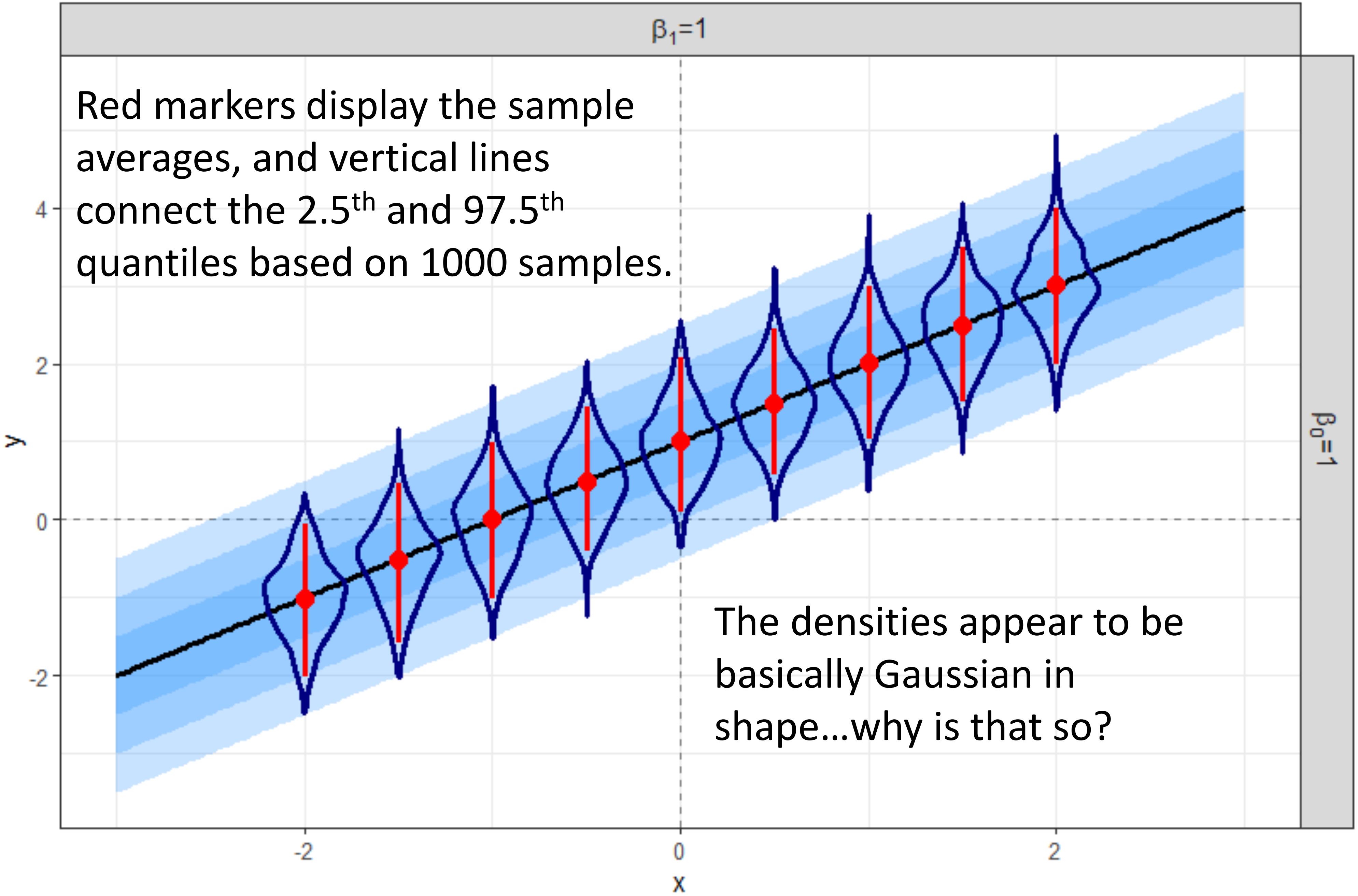
25 observations per input location…what happened to the sample averages?



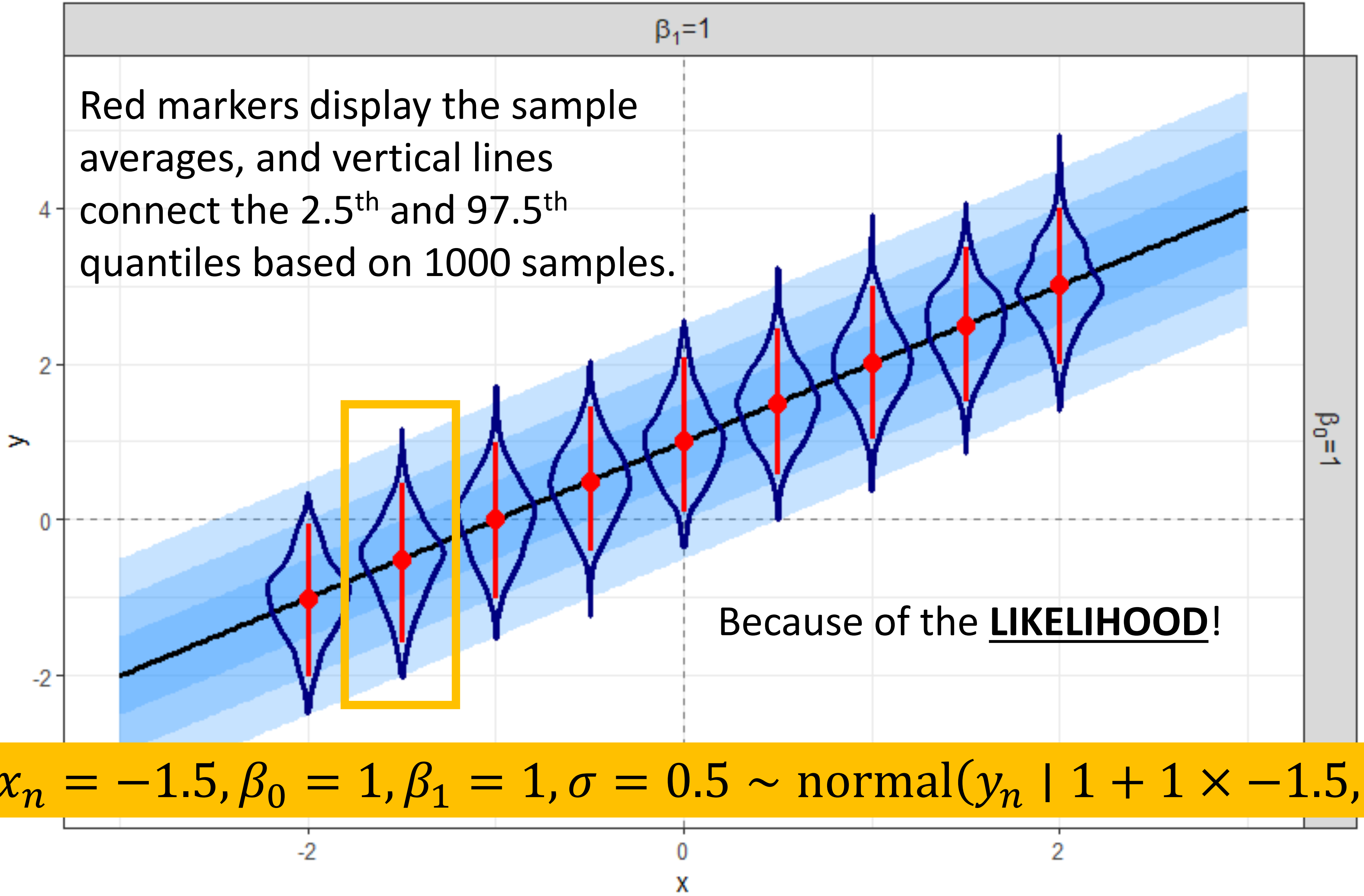
1000 observations per input location…why is the red so “wide”?



Violin plots represent the DENSITY at each input location



Violin plots represent the DENSITY at each input location

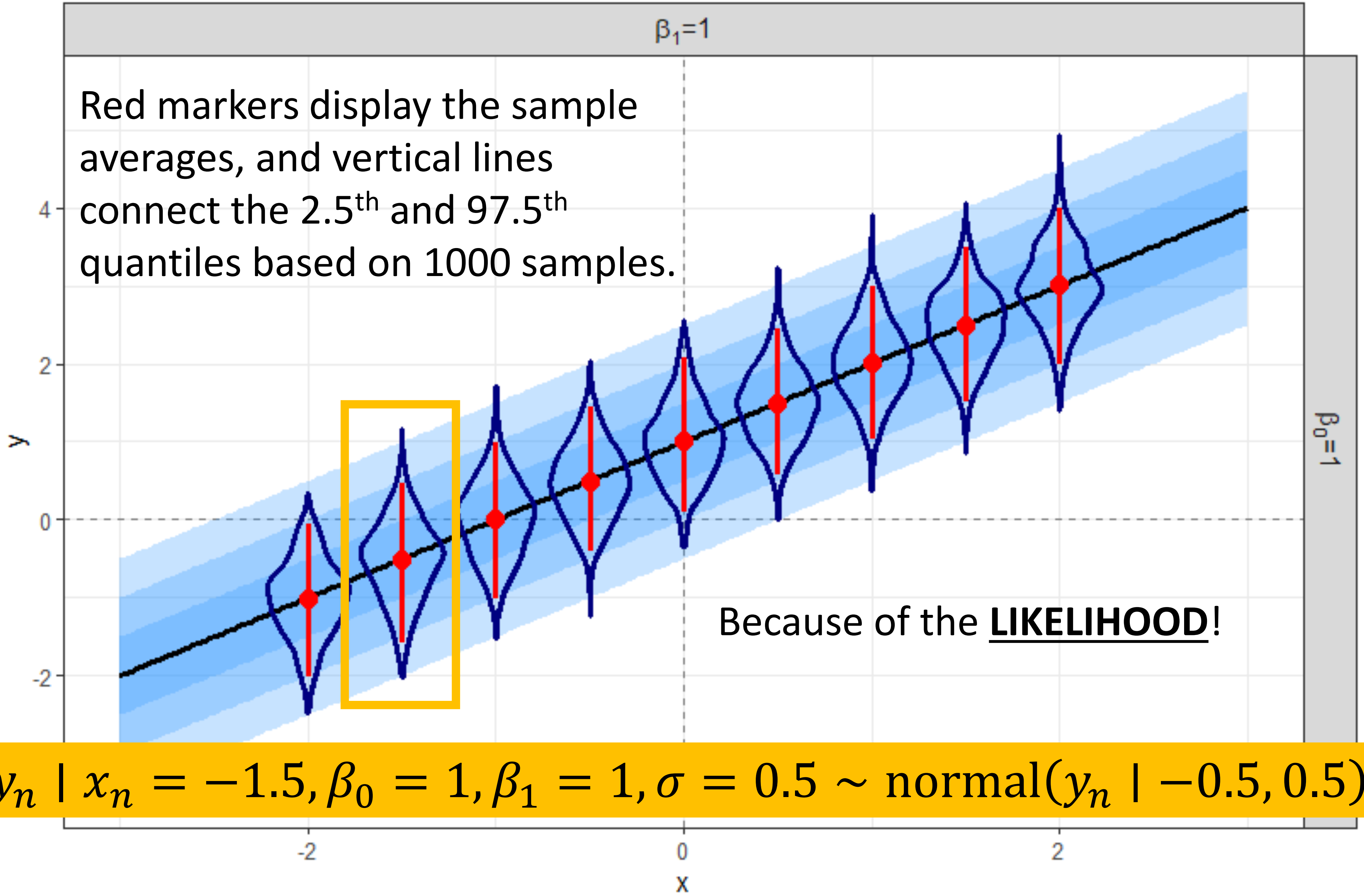


Red markers display the sample averages, and vertical lines connect the 2.5th and 97.5th quantiles based on 1000 samples.

Because of the **LIKELIHOOD**!

$$y_n \mid x_n = -1.5, \beta_0 = 1, \beta_1 = 1, \sigma = 0.5 \sim \text{normal}(y_n \mid 1 + 1 \times -1.5, 0.5)$$

Violin plots represent the DENSITY at each input location



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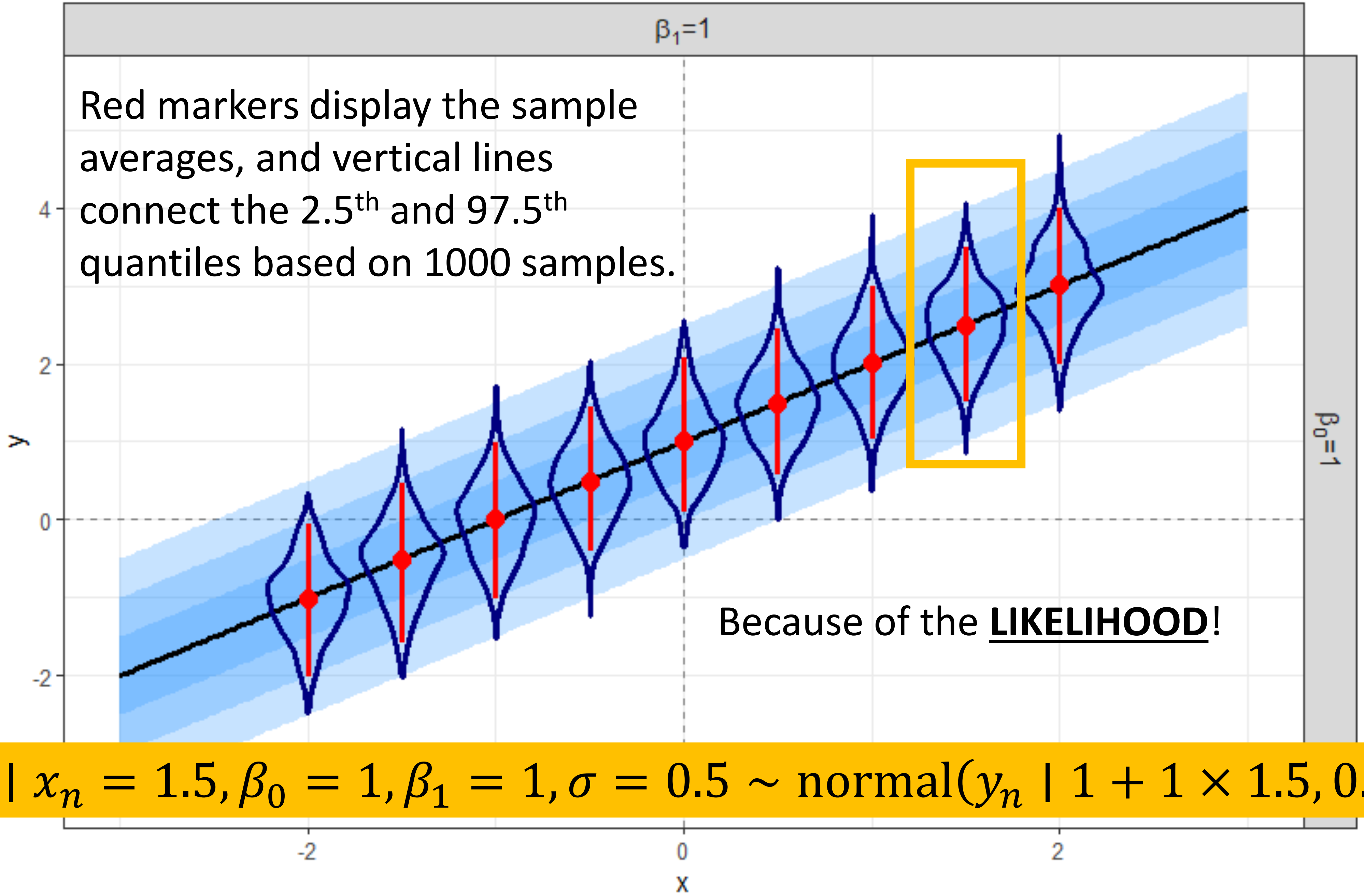
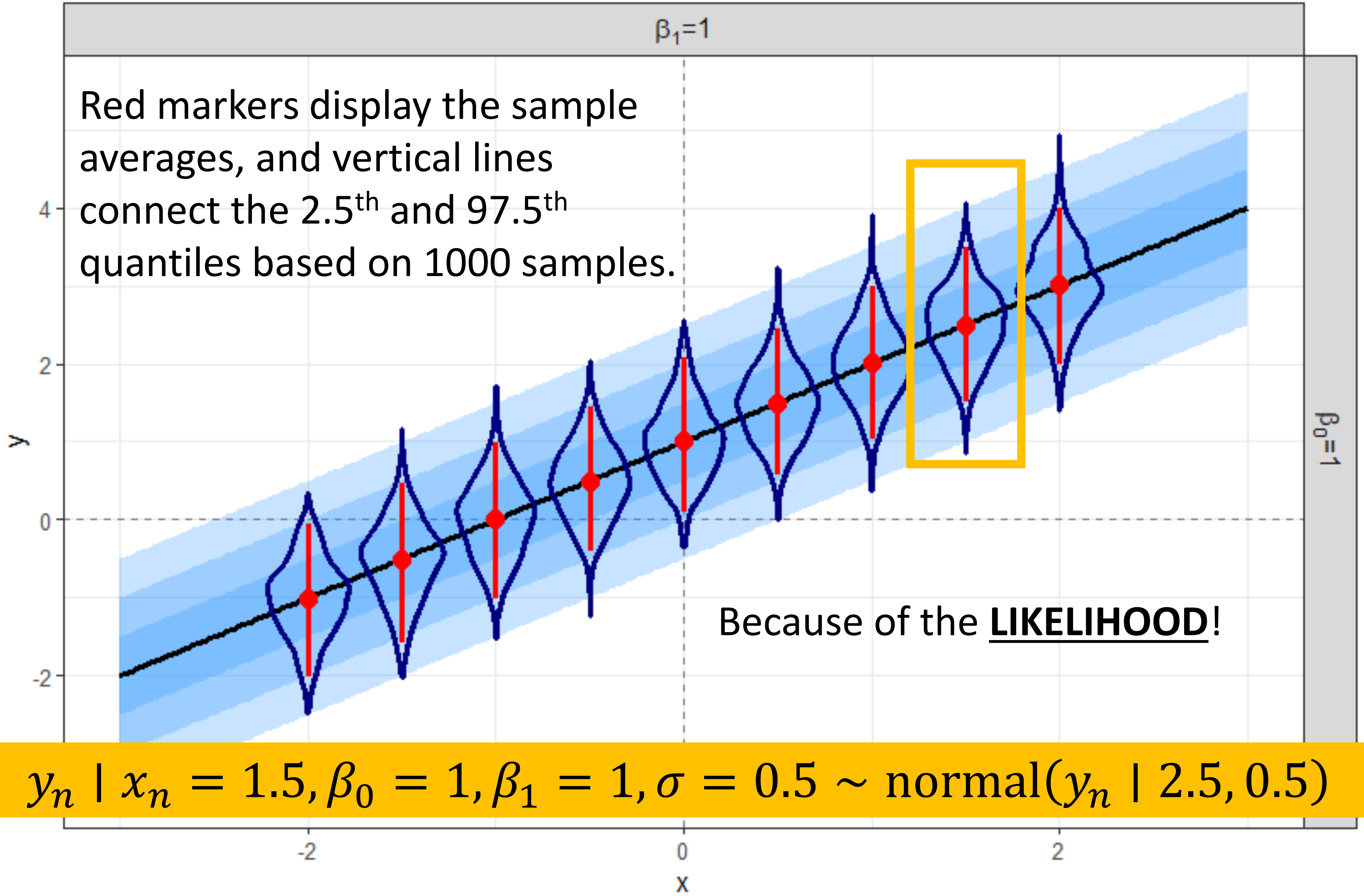


Red markers display the sample averages, and vertical lines connect the 2.5th and 97.5th quantiles based on 1000 samples.

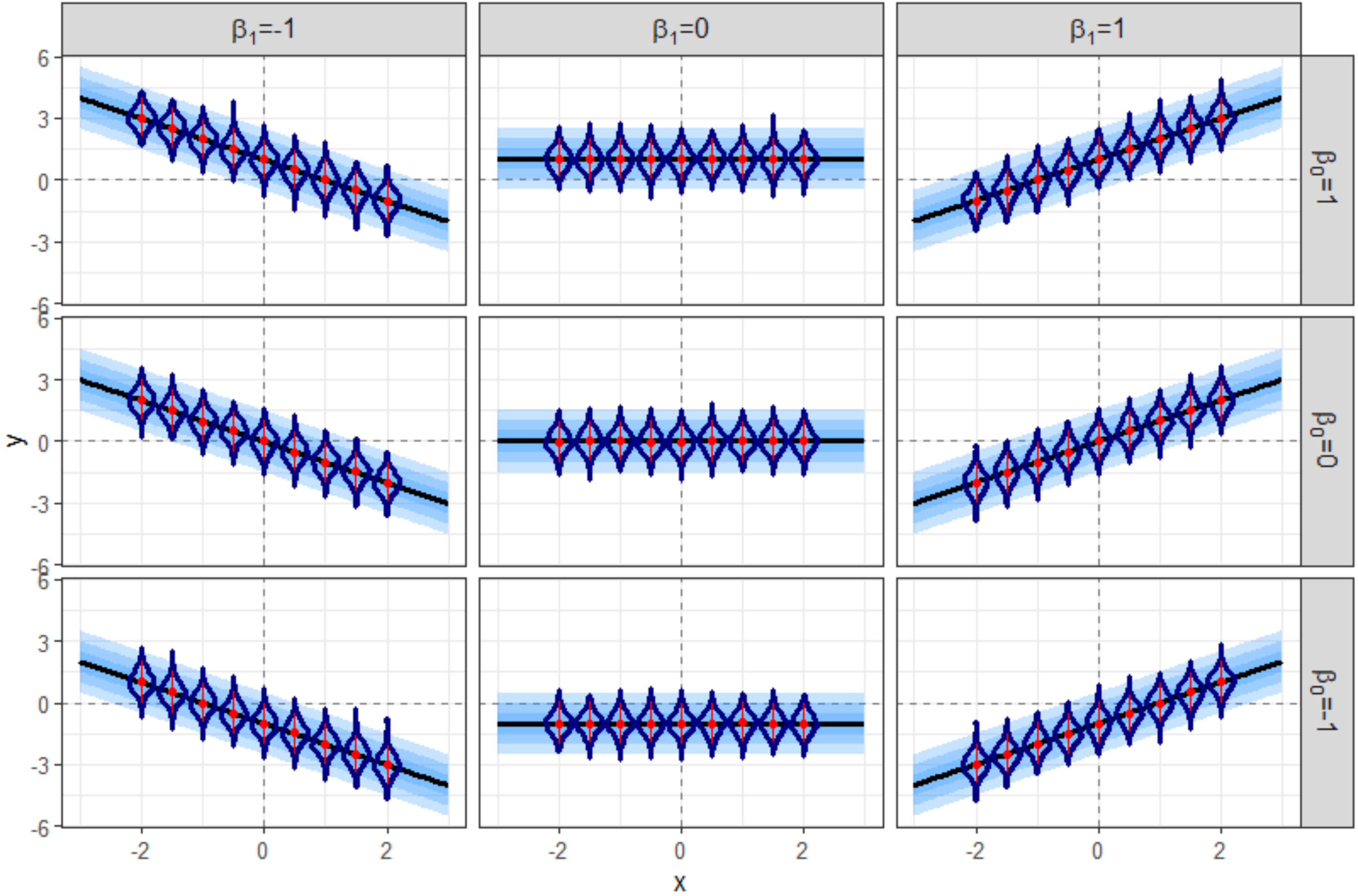
Because of the **LIKELIHOOD**!

$$y_n \mid x_n = 1.5, \beta_0 = 1, \beta_1 = 1, \sigma = 0.5 \sim \text{normal}(y_n \mid 1 + 1 \times 1.5, 0.5)$$

Violin plots represent the DENSITY at each input location



$$y_n \mid x_n = 1.5, \beta_0 = 1, \beta_1 = 1, \sigma = 0.5 \sim \text{normal}(y_n \mid 2.5, 0.5)$$



β1=-1
β1=0
β1=1
β0=1
β0=0
β0=-1
6
3
0
-3
-6
-2
0
2
y
x

Conceptual introduction to model fitting

We observed N input-output pairs. We will continue to use a linear relationship between the mean trend and the input.

- Start out assuming that σ is known.

- We need to learn the unknown intercept and the unknown slope

$$\mu_n = \beta_0 + \beta_1 x_n$$

- How can we do that? What values would be “best”?

Find the parameters that maximize the likelihood!

$$p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma) = \prod_{n=1}^{N} \{p(y_n \mid \mu_n, \sigma)\} = \prod_{n=1}^{N} \{\mathrm{normal}(y_n \mid \mu_n, \sigma)\}$$

$$\mu_n = \beta_0 + \beta_1 x_n$$

- The learning problem is therefore an optimization problem!

$$\boldsymbol{\beta}_{MLE} = \max p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)$$

Find the parameters that maximize the likelihood!

$$p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma) = \prod_{n=1}^{N} \{p(y_n \mid \mu_n, \sigma)\} = \prod_{n=1}^{N} \{\mathrm{normal}(y_n \mid \mu_n, \sigma)\}$$

$$\mu_n = \beta_0 + \beta_1 x_n$$

- Rather than maximize the likelihood directly, we will maximize the log-likelihood

$$\boldsymbol{\beta}_{MLE} = \max \log[p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)]$$

First, consider the log-likelihood in terms of the mean trend μ_n

- The log-likelihood is the sum of N conditionally independent Gaussian log-likelihoods:

$$\log[p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)] = \sum_{n=1}^{N} \{\log[\mathrm{normal}(y_n \mid \mu_n, \sigma)]\}$$

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- The log-likelihood is the sum of N conditionally independent Gaussian log-likelihoods:

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$$\Downarrow$$

$$\log[p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)] = \sum_{n=1}^{N} \left\{-\frac{1}{2}\log[\sigma^2] - \frac{1}{2}\log[2\pi] - \frac{1}{2\sigma^2}(y_n - \mu_n)^2\right\}$$

Drop the constants

$$\log[p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)] \propto -\frac{1}{2\sigma^2}\sum_{n=1}^{N}\{(y_n - \mu_n)^2\}$$

This should look familiar!!!!

Drop the constants

$$\log[p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)] \propto -\frac{1}{2\sigma^2}\sum_{n=1}^{N}\{(y_n - \boxed{\mu_n})^2\}$$

The mean is NOT constant.
The mean changes as the input changes that’s why the subscript n is included.

Drop the constants

$$\log[p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)] \propto -\frac{1}{2\sigma^2}\sum_{n=1}^{N}\boxed{\{(y_n - \mu_n)^2\}}$$

What's this??

Drop the constants

$$\log[p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)] \propto -\frac{1}{2\sigma^2} \sum_{n=1}^{N} \boxed{\{(y_n - \mu_n)^2\}}$$

Substitute in the expression for the mean trend

$$y_n - \mu_n = y_n - (\beta_0 + \beta_1 x_n) = \epsilon_n$$

The n-th observation's error or residual!!

Drop the constants

$$\log[p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)] \propto -\frac{1}{2\sigma^2}\sum_{n=1}^{N}\boxed{\{(y_n - \mu_n)^2\}}$$

Substitute in the expression for the mean trend

Square the residual:

$$(y_n - \mu_n)^2 = \epsilon_n^2$$

The summation term is the **residual sum of squares (RSS)** or the **sum of squared errors (SSE)**!

$$RSS = SSE = \sum_{n=1}^{N} \{(y_n - \mu_n)^2\} = \sum_{n=1}^{N} \{\epsilon_n^2\}$$

- The log-likelihood is therefore proportional to the negative of the SSE:

$$\log[p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)] \propto -SSE$$

The summation term is the **residual sum of squares (RSS)** or the **sum of squared errors (SSE)**!

$$\log[p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)] \propto -SSE$$

- Maximizing the (log) likelihood is therefore equivalent to minimizing the squared error!

The summation term is the **residual sum of squares (RSS)** or the **sum of squared errors (SSE)**!

$$\log[p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)] \propto -SSE$$

- Maximizing the (log) likelihood is therefore equivalent to minimizing the squared error!

The Maximum Likelihood Estimates (MLEs) to the linear model parameters will be equivalent to the Ordinary Least Squares (OLS) estimates!

Bayesian formulation

- Still consider σ to be known.
- We need to specify our prior belief about the unknown $\boldsymbol{\beta}$'s.
- We update our prior based on the observations, to yield the posterior on the unknowns.

Bayesian formulation

$$p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{x}, \sigma) \propto p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma) \times p(\boldsymbol{\beta})$$

Bayesian formulation

$$p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{x}, \sigma) \propto p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma) \times p(\boldsymbol{\beta})$$

- If we would use a diffuse, uninformative, or ignorant prior on $\boldsymbol{\beta}$: $p(\boldsymbol{\beta}) \propto 1$…

- How does the posterior behave?

Bayesian formulation with a diffuse prior follows the likelihood!

$$p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{x}, \sigma) \propto p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma) \times 1$$

$$p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{x}, \sigma) \propto p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)$$

We know that the MLE is equivalent to the OLS estimate. We should expect that with a diffuse prior the MAP should also correspond to the OLS estimate!

Bayesian formulation with an informative prior

$$p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{x}, \sigma) \propto p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma) \times p(\boldsymbol{\beta})$$

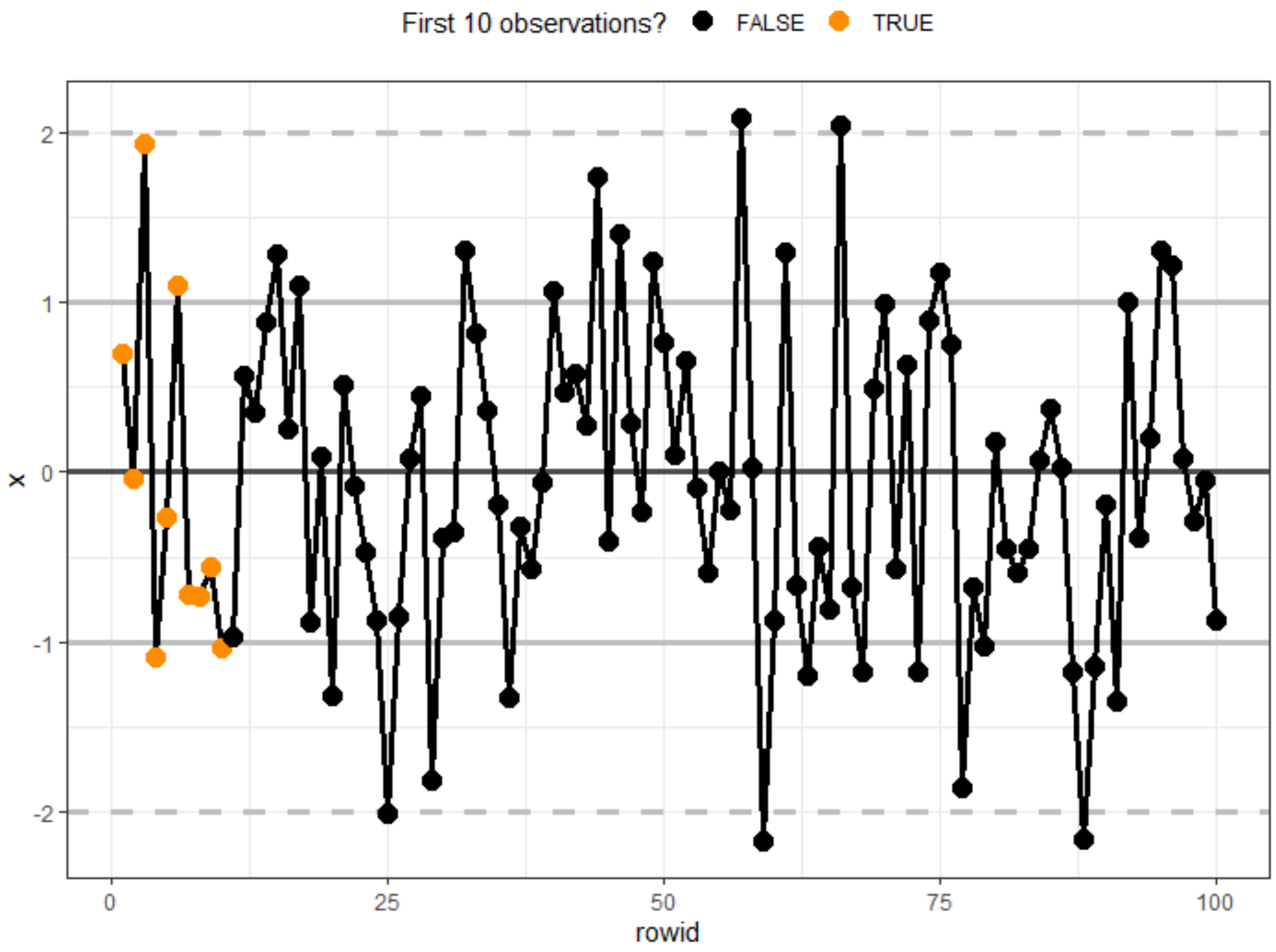
- The likelihood is still the same…it wants to minimize the error…
- What impact will the prior have?

Synthetic data example

- Let’s see if we can learn the parameters of a linear model.
- Data will be generated similar to what we did before.
- We will use different values of the intercept, slope, and noise compared to our previous visualizations.
- The input will be generated from a standard normal.

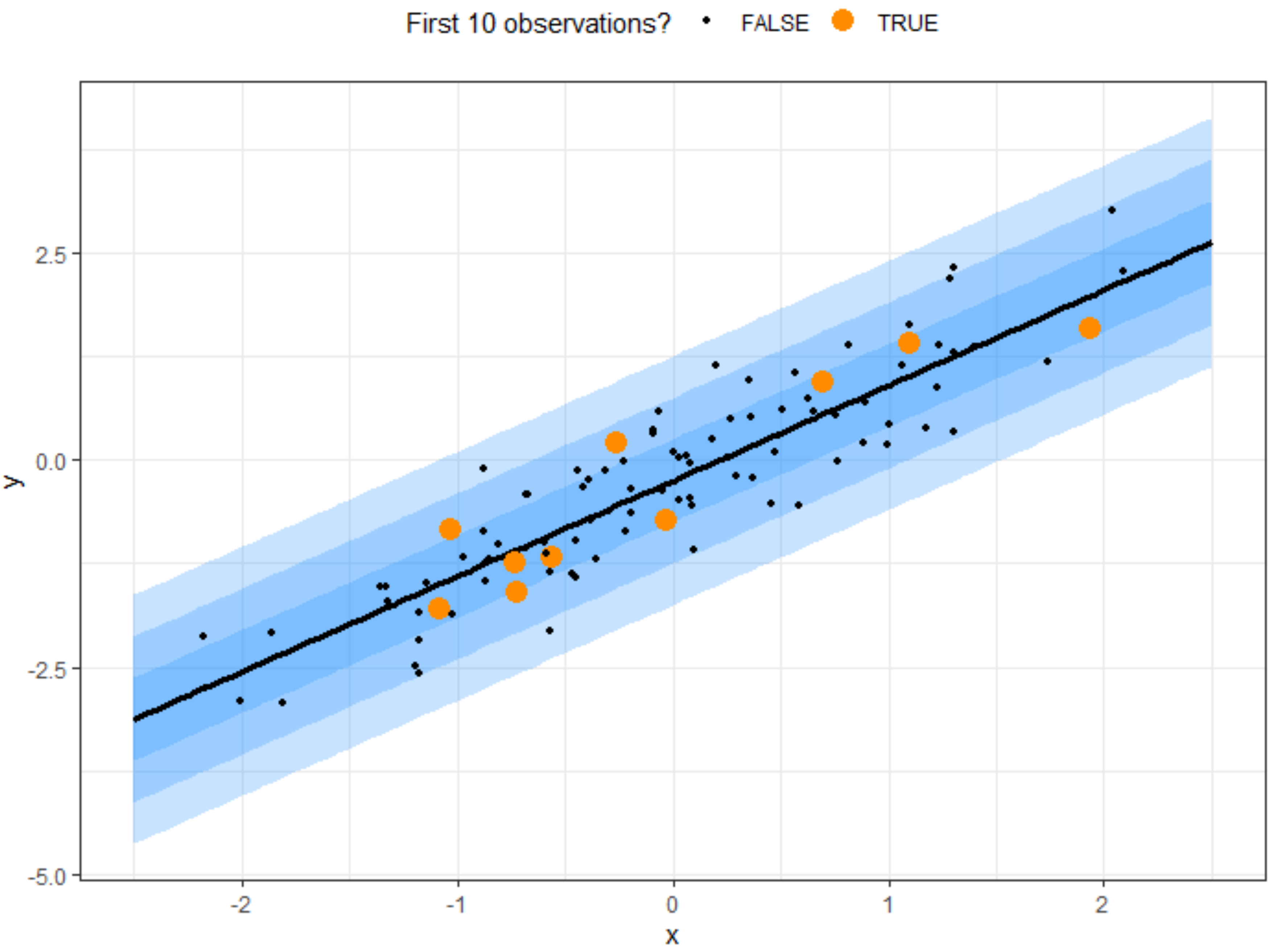
```
x <- rnorm(n = 100)
```

Generate 100 total random input values.



Specify true values for the parameters.

- Set $\beta_0 = -0.25, \beta_1 = 1.15$
- Calculate the true mean trend at each x_n.
- Set $\sigma = 0.5$
- Generate random observations based on the true mean trend and the specified noise.



First 10 observations?
FALSE
TRUE
2.5
0.0
-2.5
-5.0
y
-2
-1
0
1
2
x

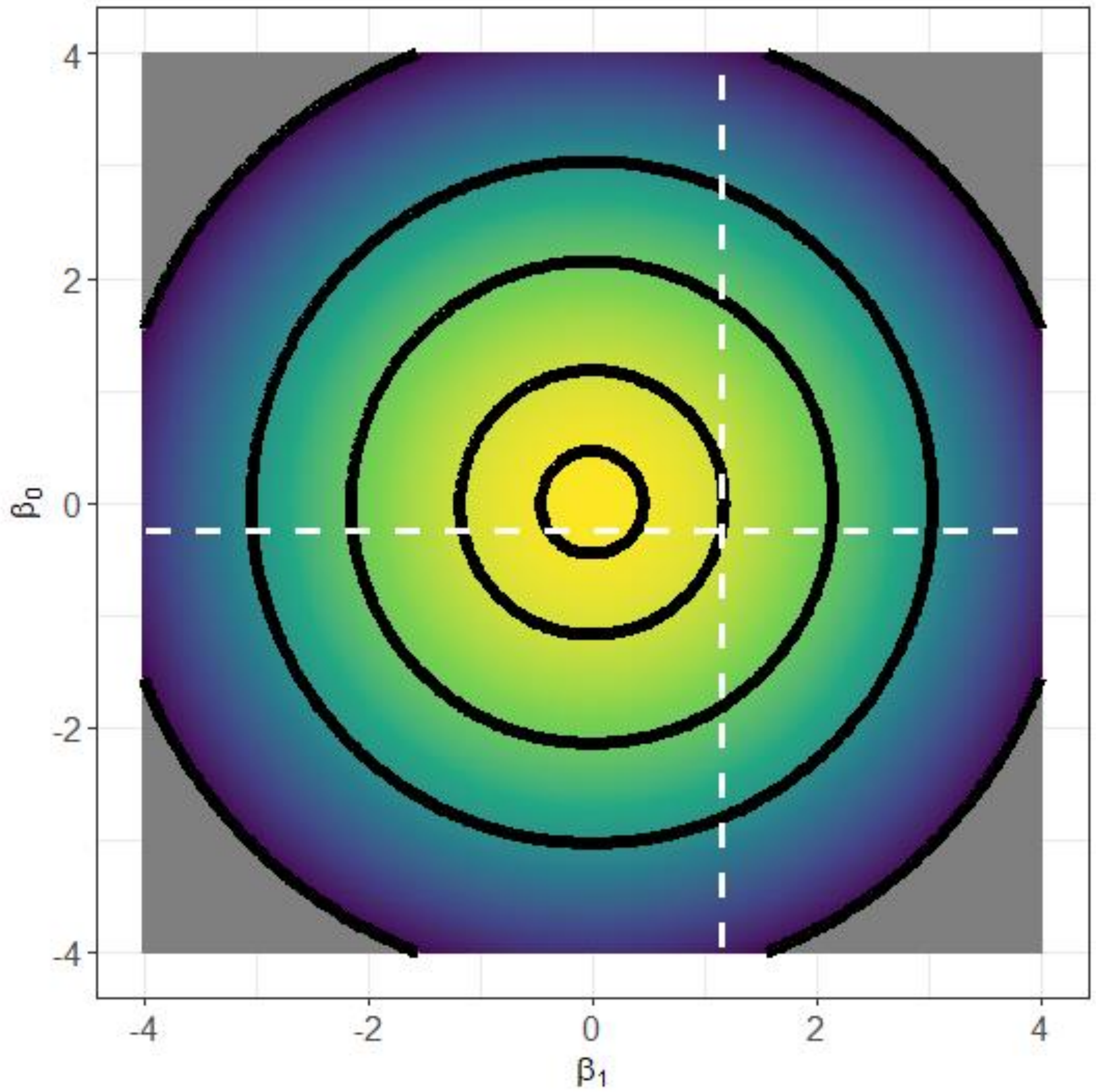
Prior specification

- Let’s assume that a-priori the parameters are independent:

$$p(\boldsymbol{\beta}) = p(\beta_0)p(\beta_1)$$

- Let’s assign standard normal distributions to both parameters:

$$p(\beta) = \mathrm{normal}(\beta_0 \mid 0,1) \times \mathrm{normal}(\beta_1 \mid 0,1)$$


4
2
0
-2
-4
-4
-2
0
2
4
β_0
β_1

Let’s visualize the log-posterior surface based on the first 10 observations, the complete probability model is:

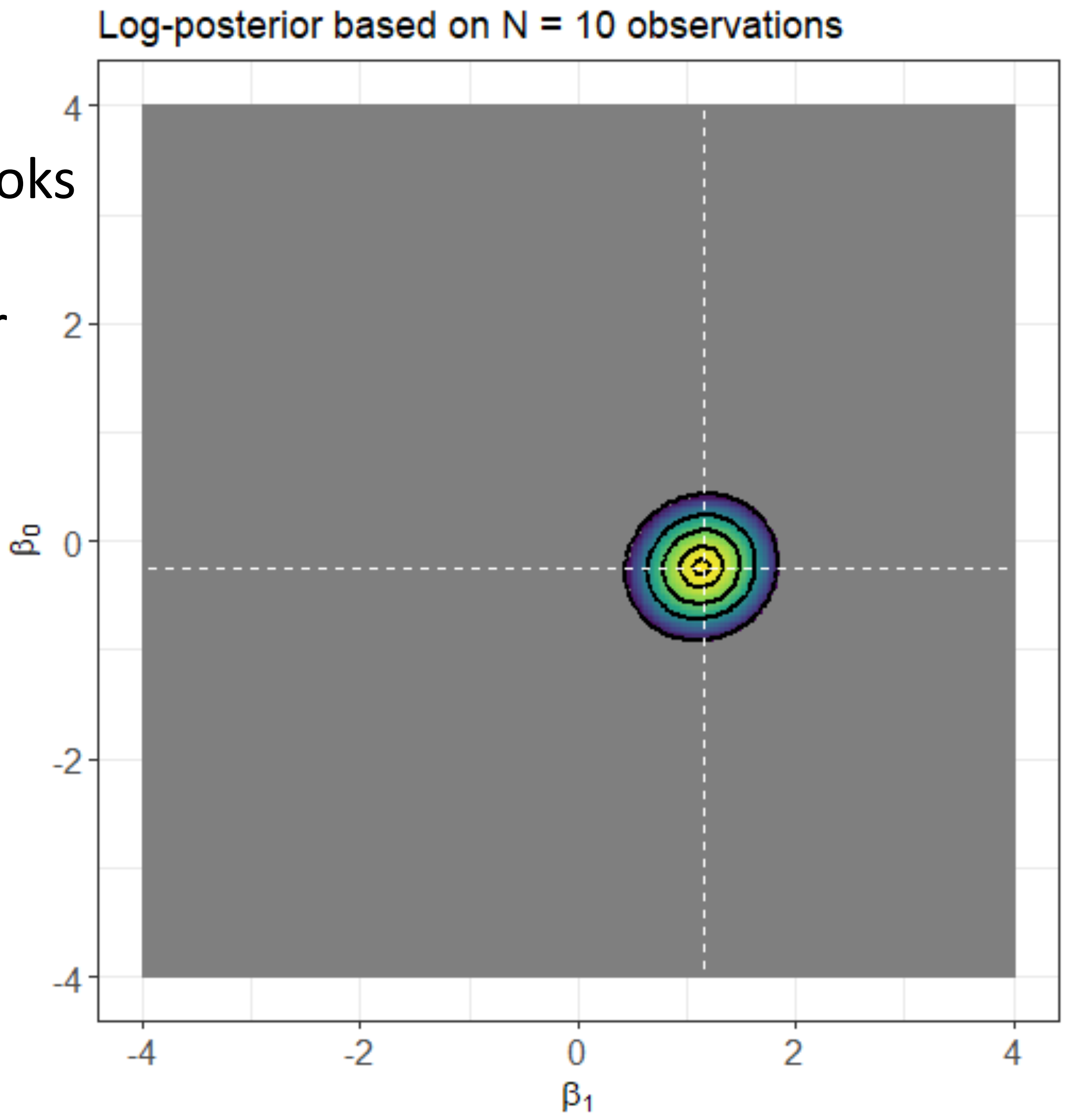
$$p(y_n \mid \mu_n, \sigma) \sim \mathrm{normal}(y_n \mid \mu_n, \sigma)$$

$$\mu_n = \beta_0 + \beta_1 x_n$$

$$\beta_0 \sim \mathrm{normal}(\beta_0 \mid 0,1)$$

$$\beta_1 \sim \mathrm{normal}(\beta_1 \mid 0,1)$$

The posterior looks centered on the TRUE parameter values!!!

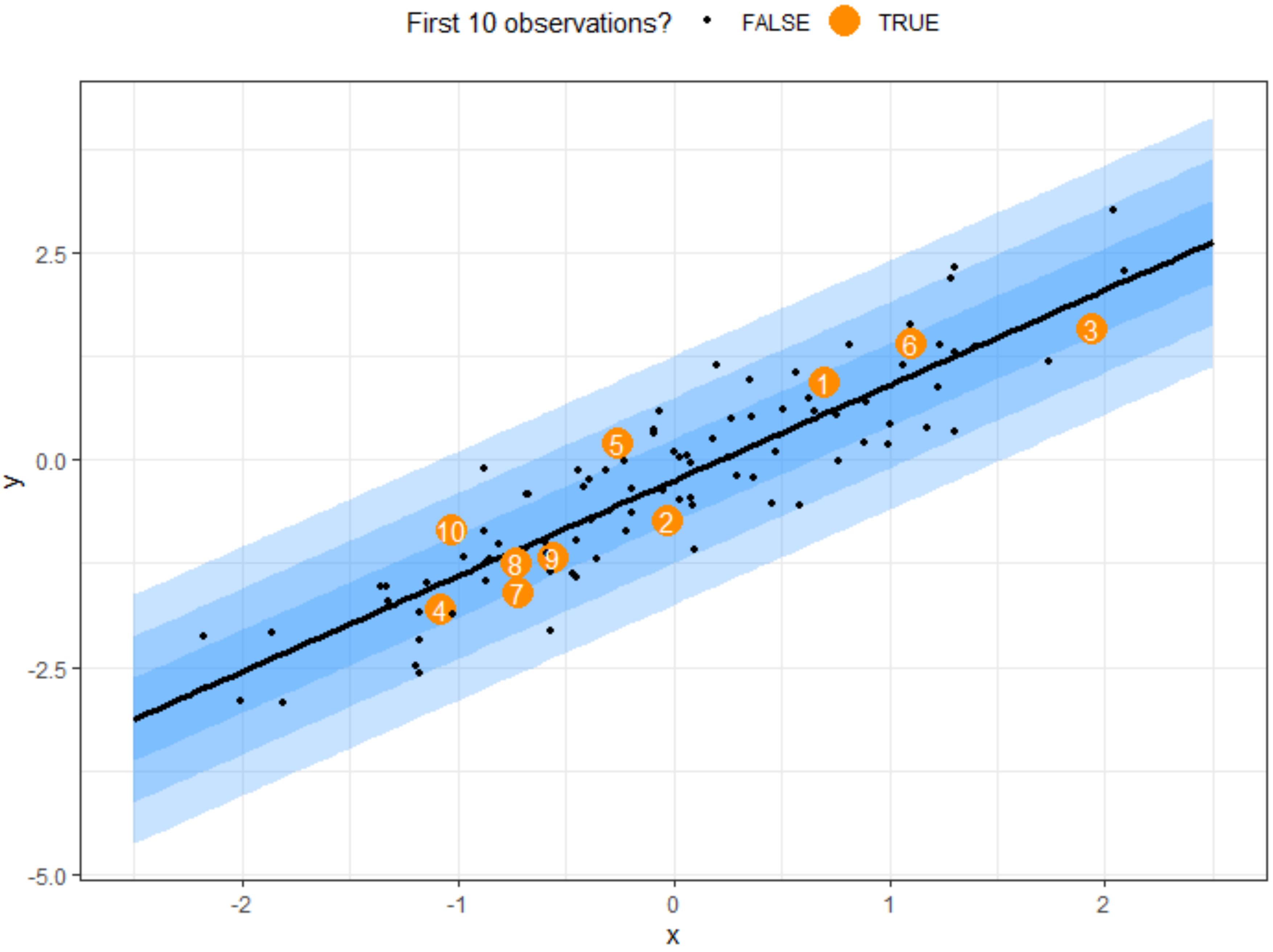


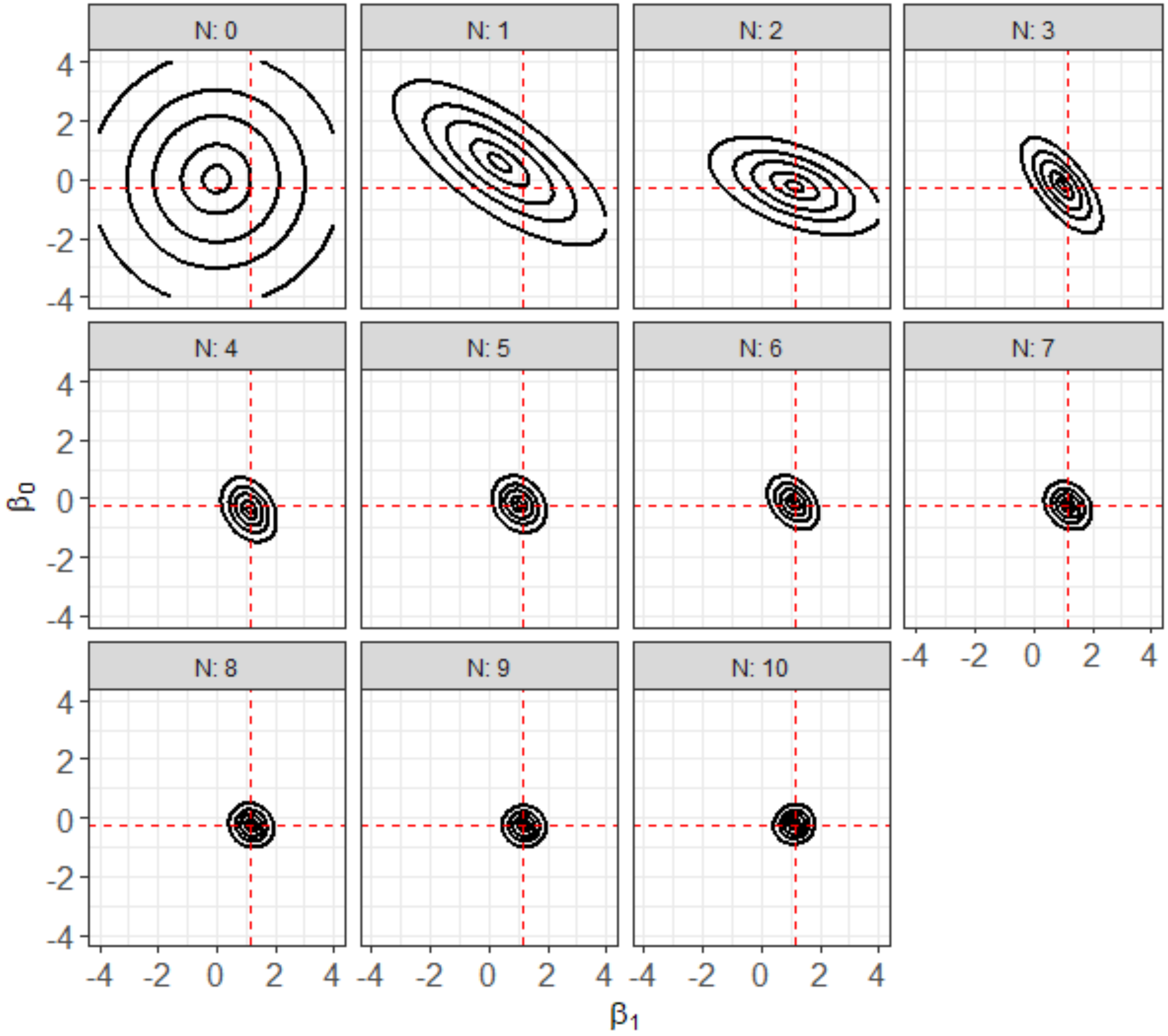
Posterior precision is VERY high!!

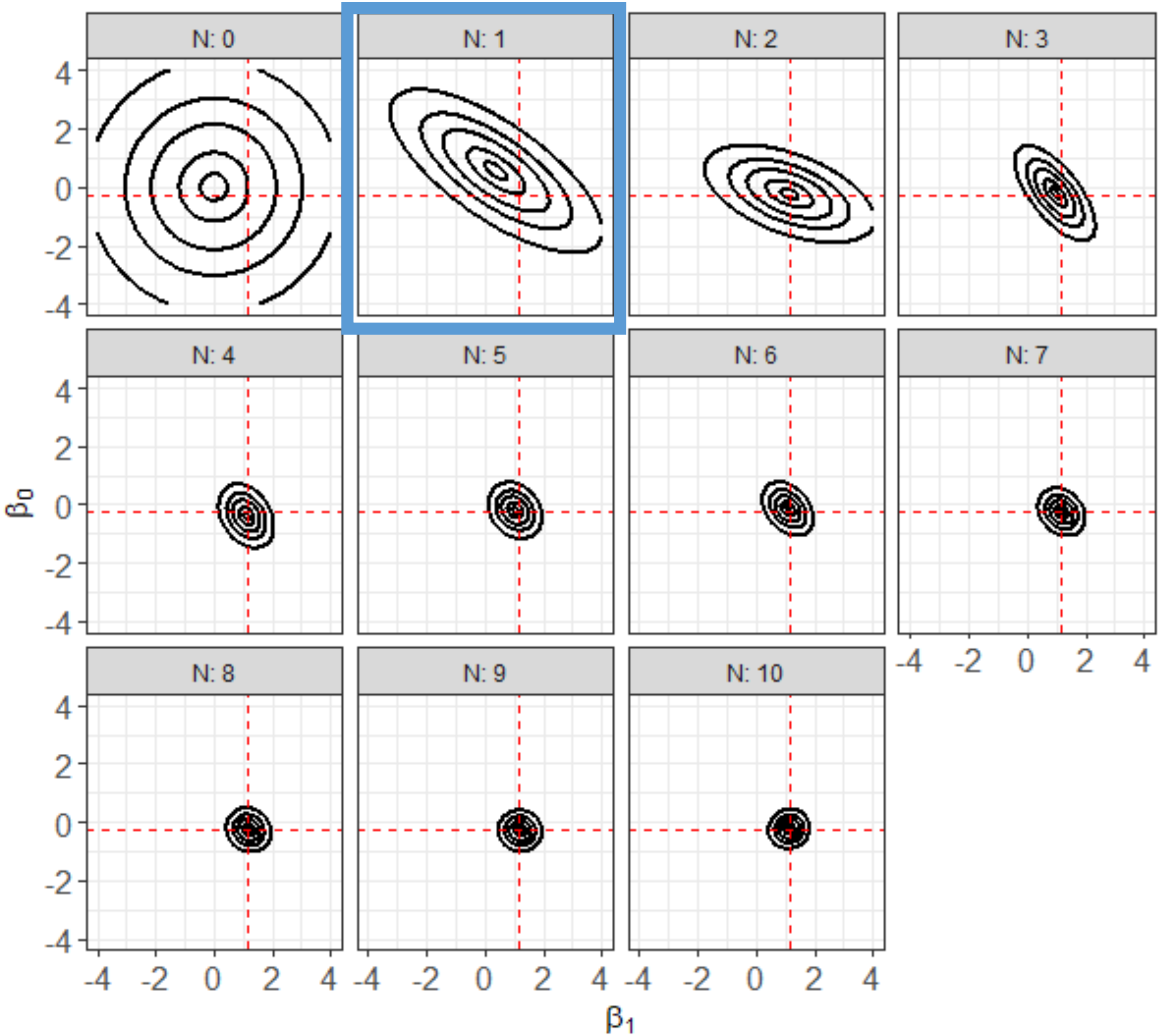
Let’s see how we got to that point.

- Sequentially add the first 10 data points.

- We will examine how the joint posterior between the unknown intercept and slope change.


First 10 observations?
FALSE
TRUE
2.5
0.0
-2.5
-5.0
y
-2
-1
0
1
2
x
1
2
3
4
5
6
7
8
9
10

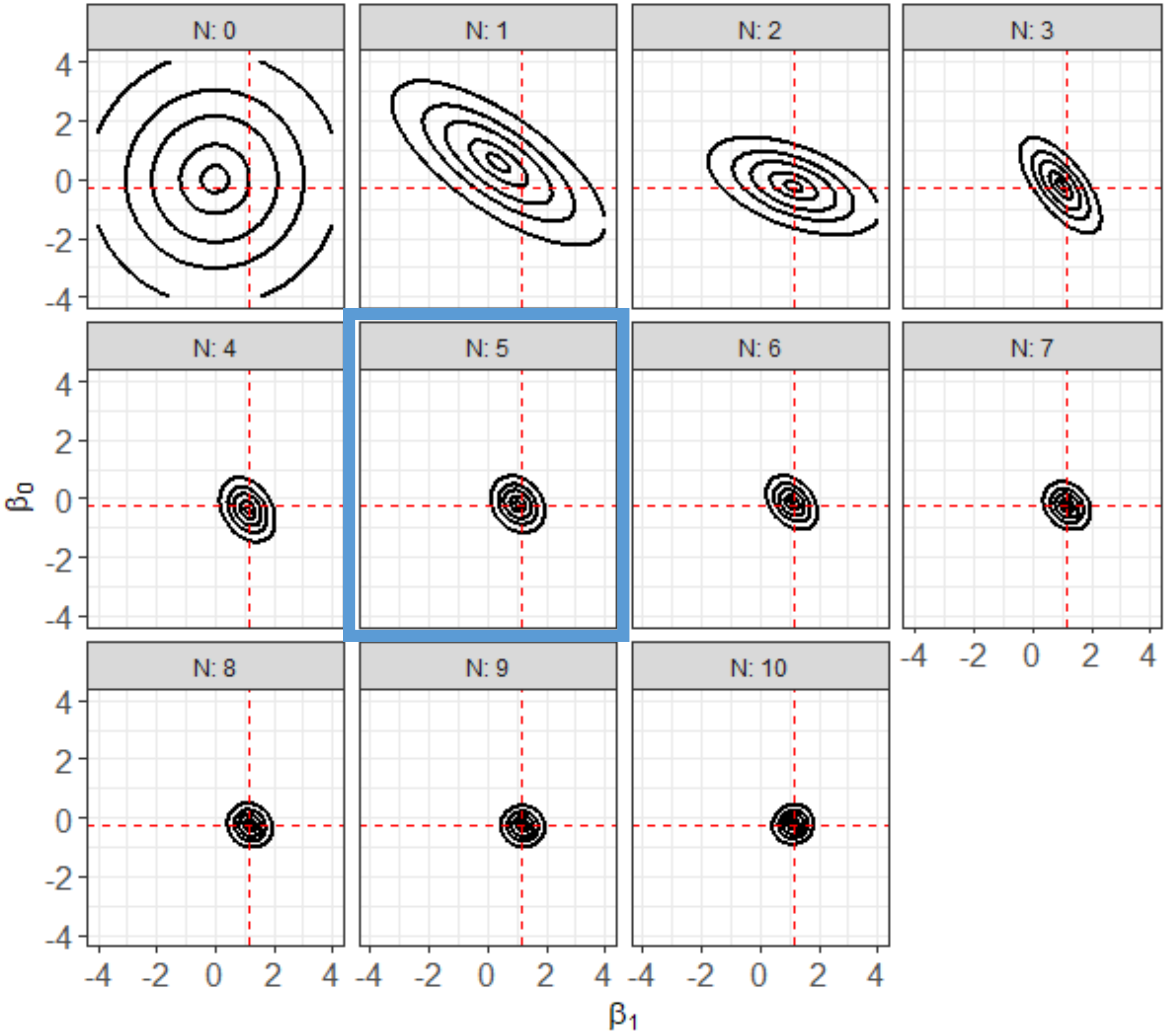

N: 0
N: 1
N: 2
N: 3
N: 4
N: 5
N: 6
N: 7
N: 8
N: 9
N: 10
β_0
β_1
4
2
0
-2
-4



After just a single observation…we have updated our belief about 2 unknowns!!!!

The prior provides information just like when we were learning an unknown constant mean!

Are the parameters independent after 1 observation?



The posterior seems relatively unchanged after 5 observations…

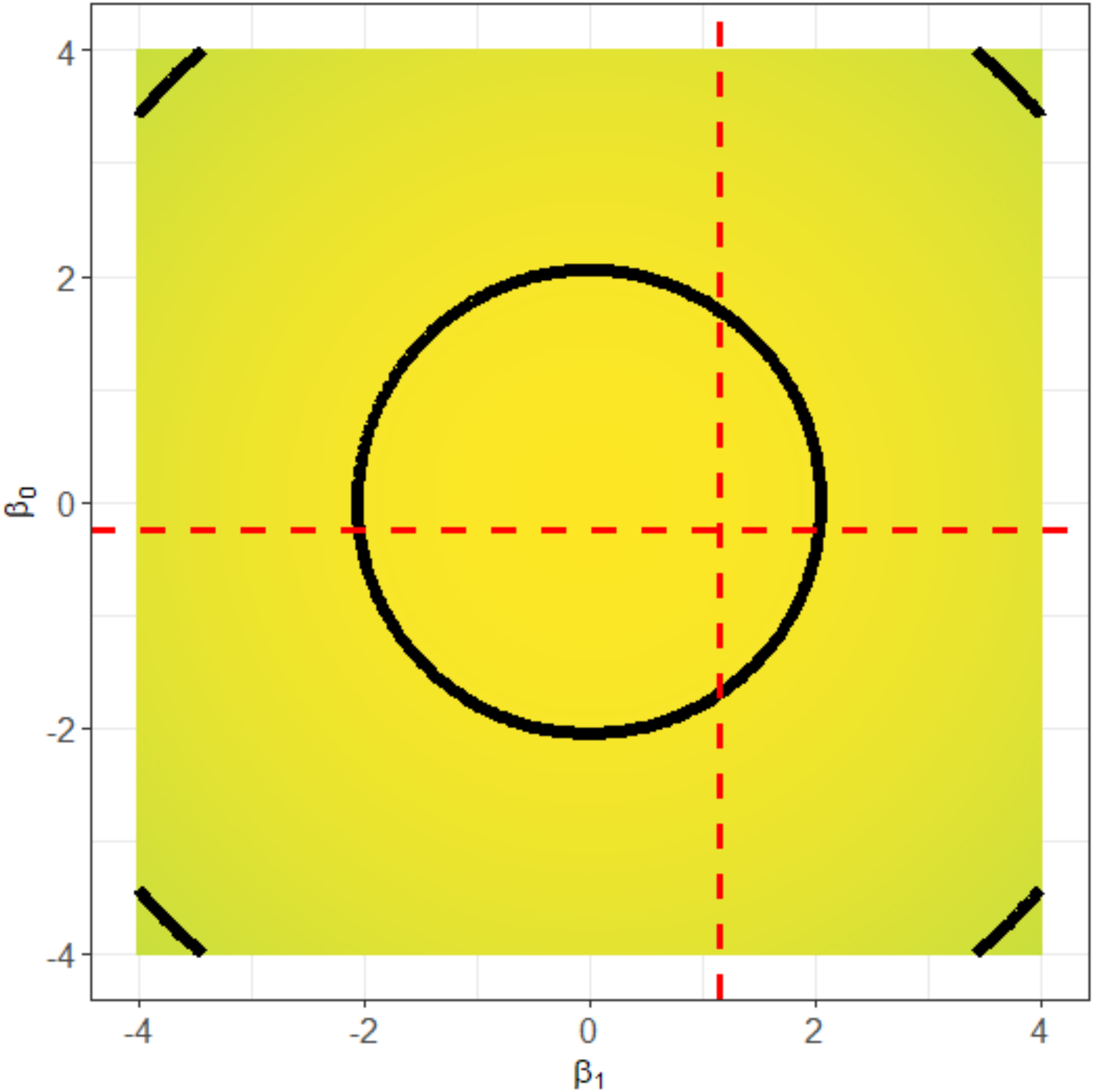
Could the prior be influencing the posterior too much?

Try again, but this time with a diffuse, uninformative, or ignorant prior

- Continue to assume the parameters are a-prior independent.

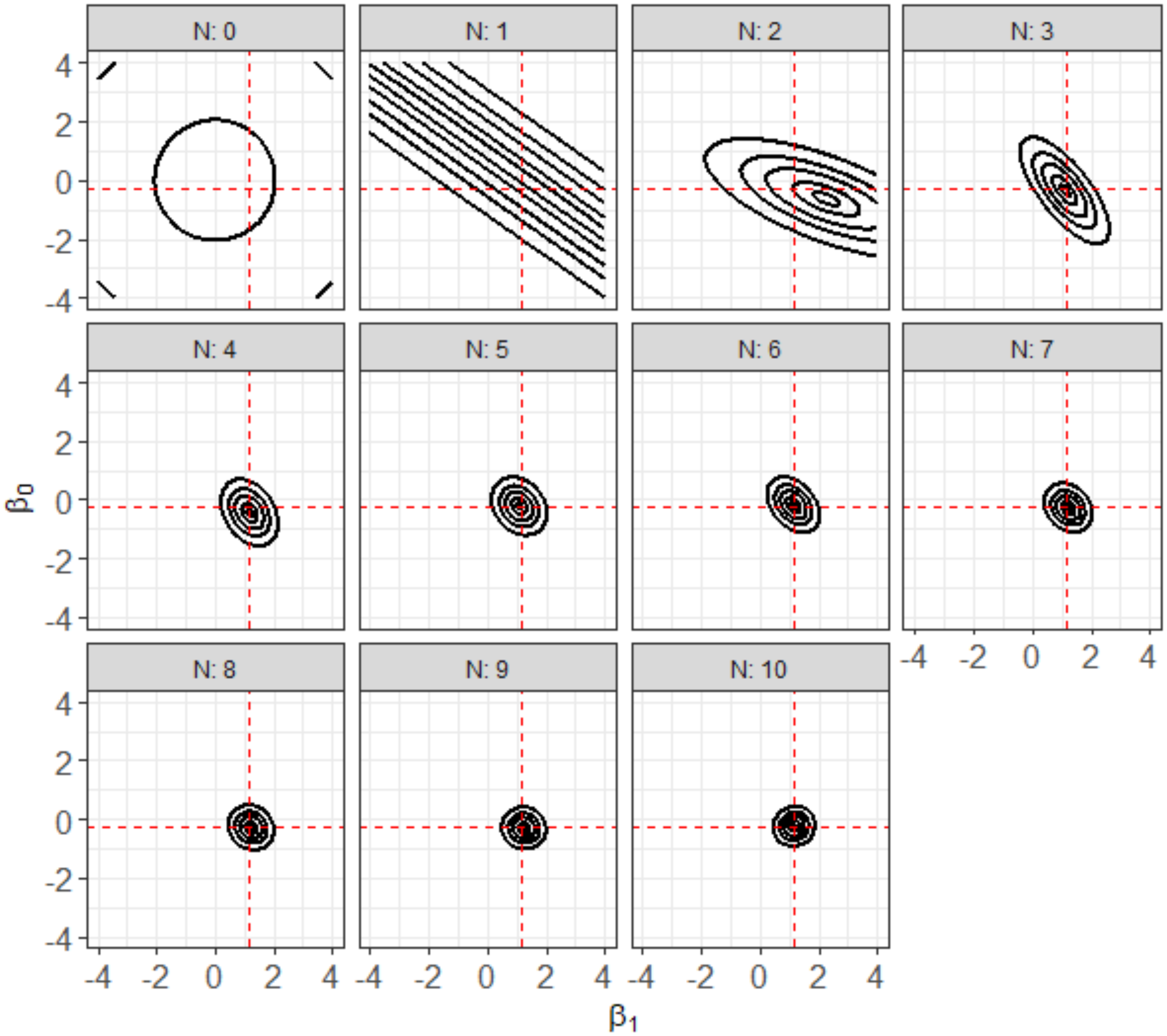
- But increase the prior standard deviations from 1 to 20.

$$p(\beta) = \text{normal}(\beta_0 \mid 0{,}20) \times \text{normal}(\beta_1 \mid 0{,}20)$$

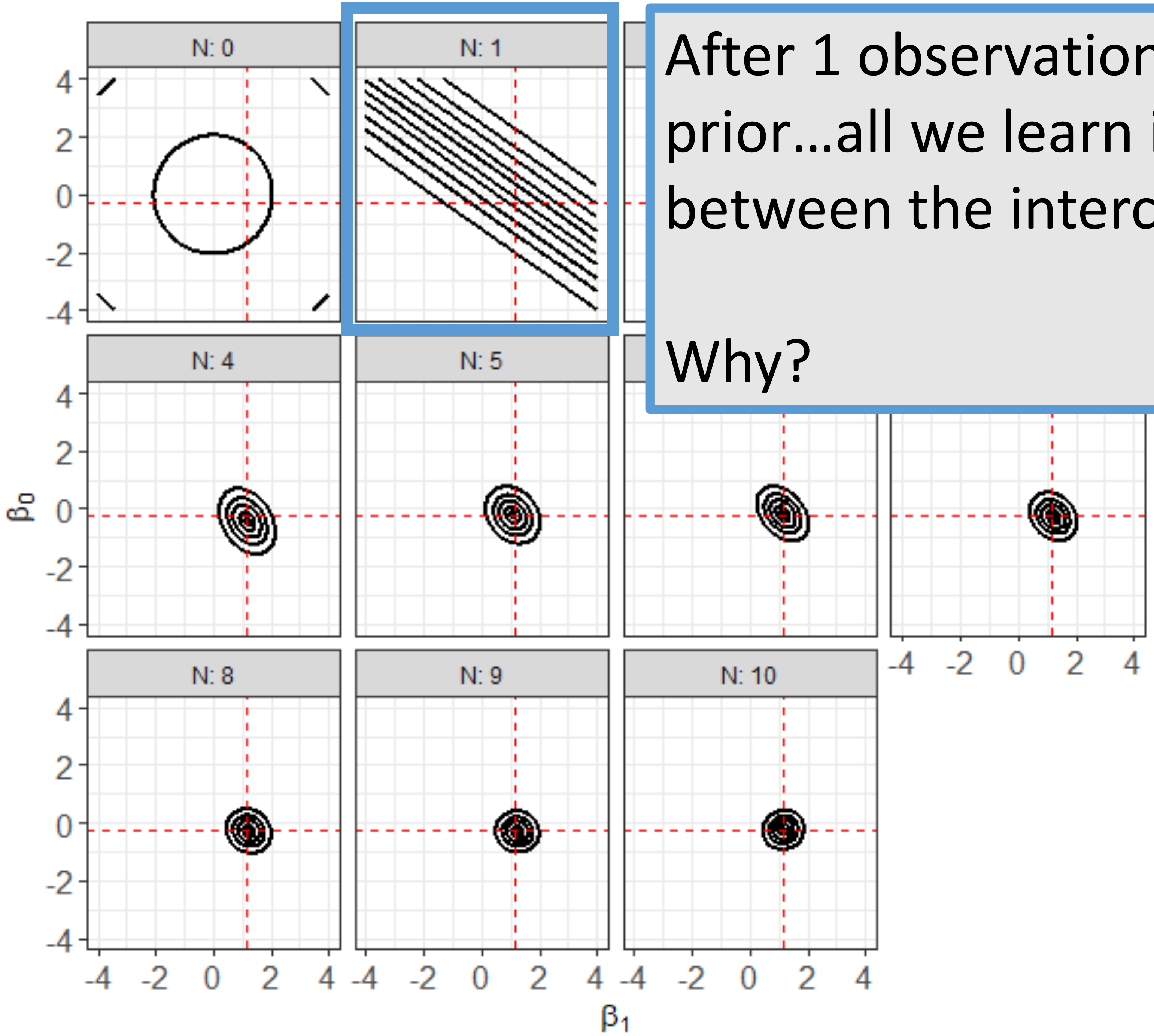

4
2
0
-2
-4
β0
-4
-2
0
2
4
β1

Repeat evaluating the log-posterior by sequentially adding the first 10 observations

- Using such a diffuse prior forces the posterior to follow the likelihood.

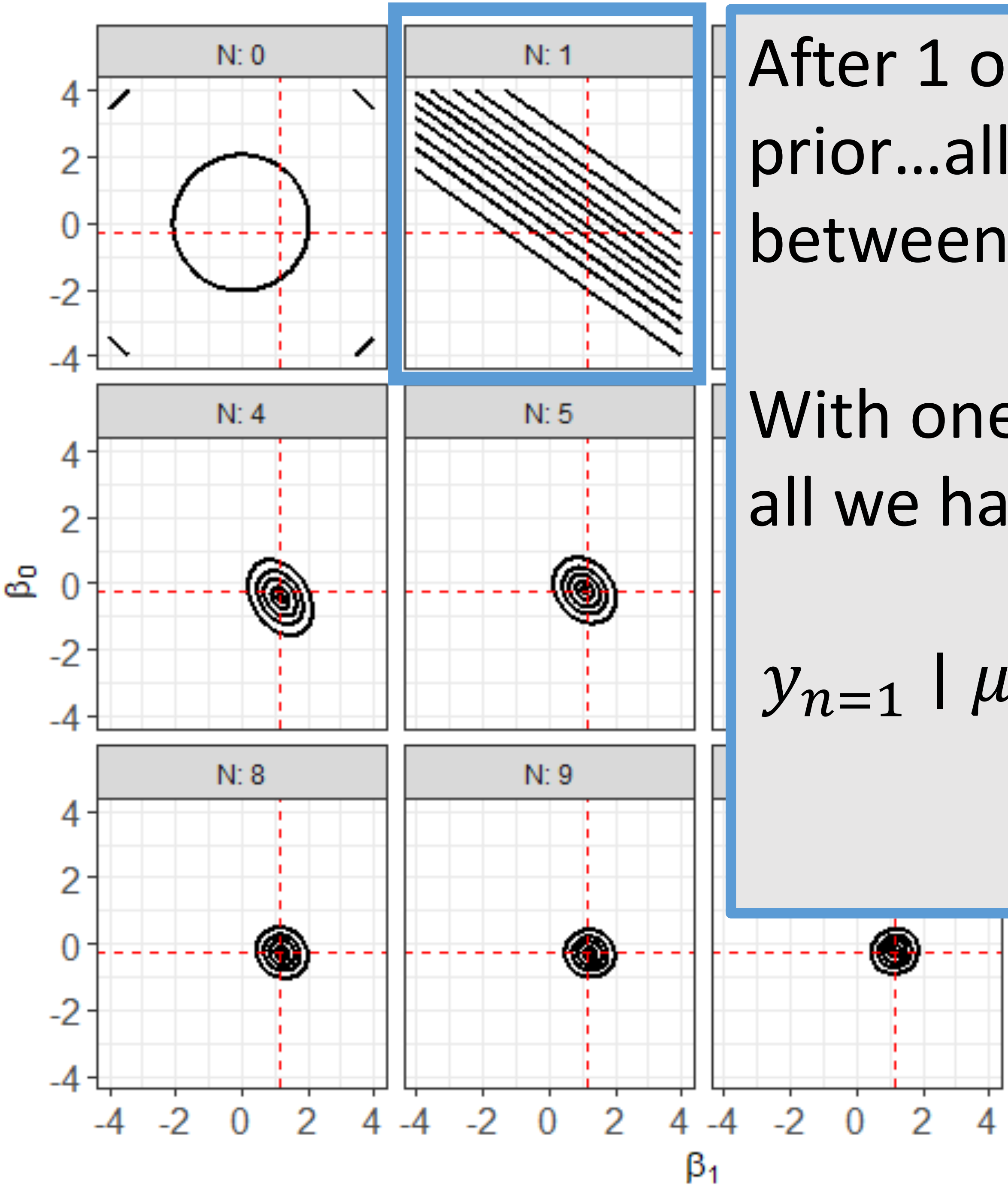


N: 0
N: 1
N: 2
N: 3
N: 4
N: 5
N: 6
N: 7
N: 8
N: 9
N: 10
β_0
β_1
4
2
0
-2
-4



After 1 observation with a diffuse prior…all we learn is a TRADE-OFF between the intercept and the slope!

Why?

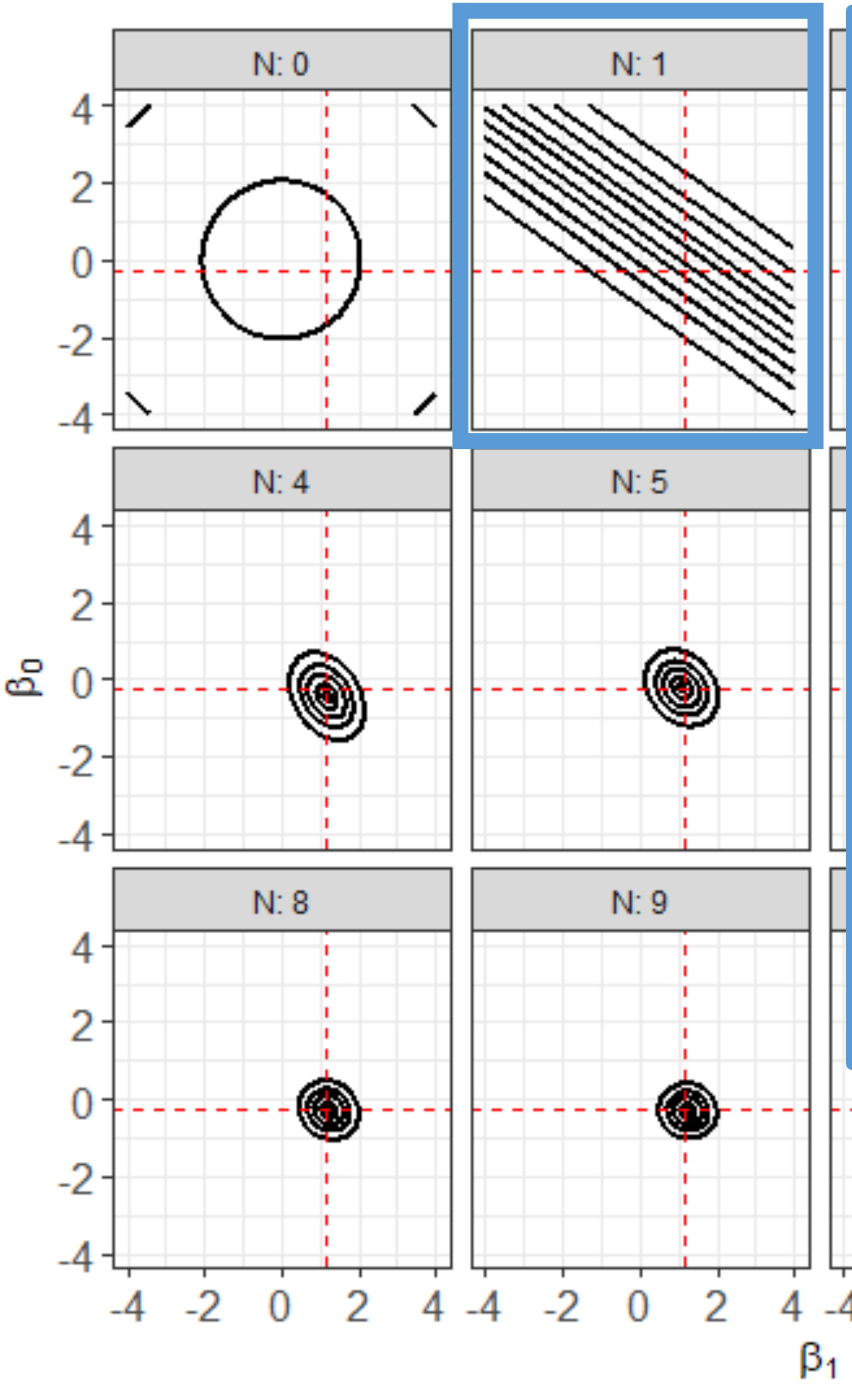


After 1 observation with a diffuse prior…all we learn is a TRADE-OFF between the intercept and the slope!

With one observation and a diffuse prior, all we have to consider is the likelihood:

$$y_{n=1} \mid \mu_{n=1}, \sigma \sim \mathrm{normal}(y_{n=1} \mid \mu_{n=1}, \sigma)$$

$$\mu_{n=1} = \beta_0 + \beta_1 x_{n=1}$$



After 1 observation with a diffuse prior…all we learn is a TRADE-OFF between the intercept and the slope!

With one observation and a diffuse prior, all we have to consider is the likelihood:

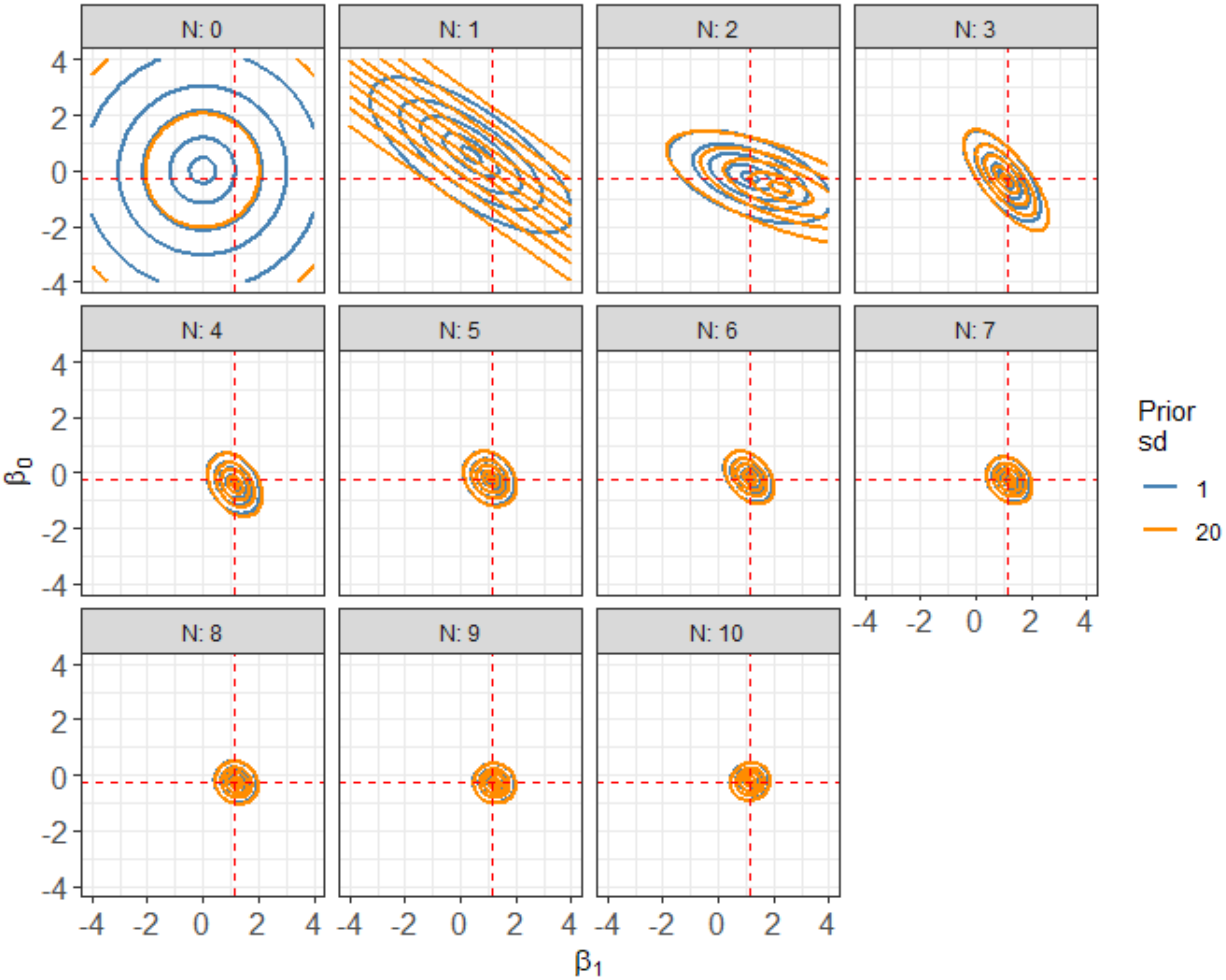
$$y_{n=1} \mid \mu_{n=1}, \sigma \sim \text{normal}(y_{n=1} \mid \mu_{n=1}, \sigma)$$

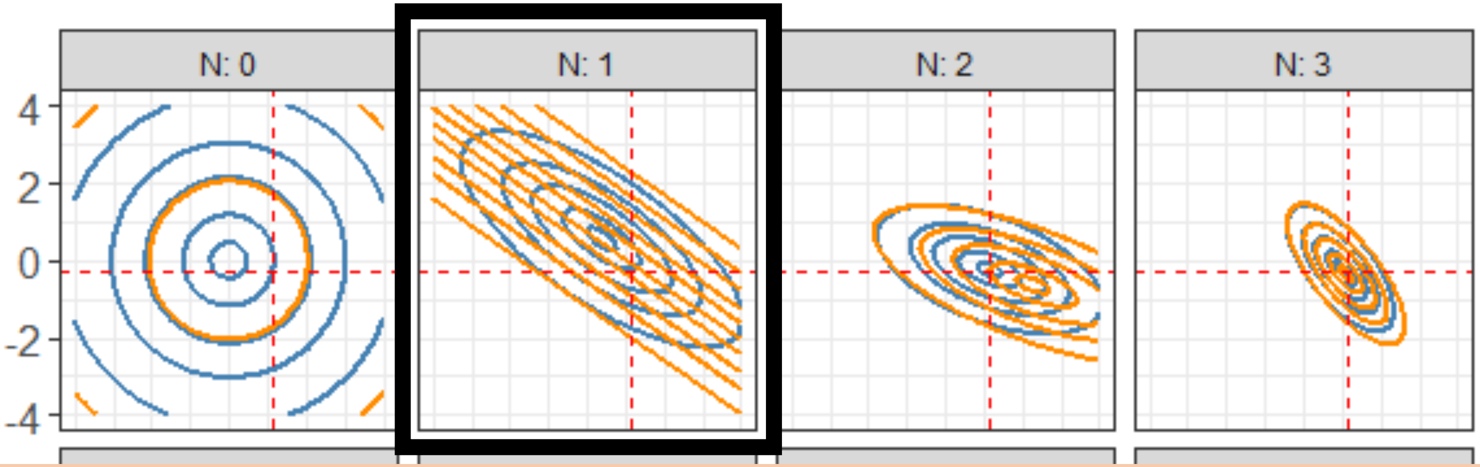
constant =

$$\mu_{n=1} = \beta_0 + \beta_1 x_{n=1}$$

Increasing the slope, while decreasing the intercept will yield a constant value for $\mu_{n=1}$. The plotted contours represent that trade-off!

Directly compare the posterior based on the two priors as we add points.


N: 0
N: 1
N: 2
N: 3
N: 4
N: 5
N: 6
N: 7
N: 8
N: 9
N: 10
β_0
β_1
Prior sd
1
20



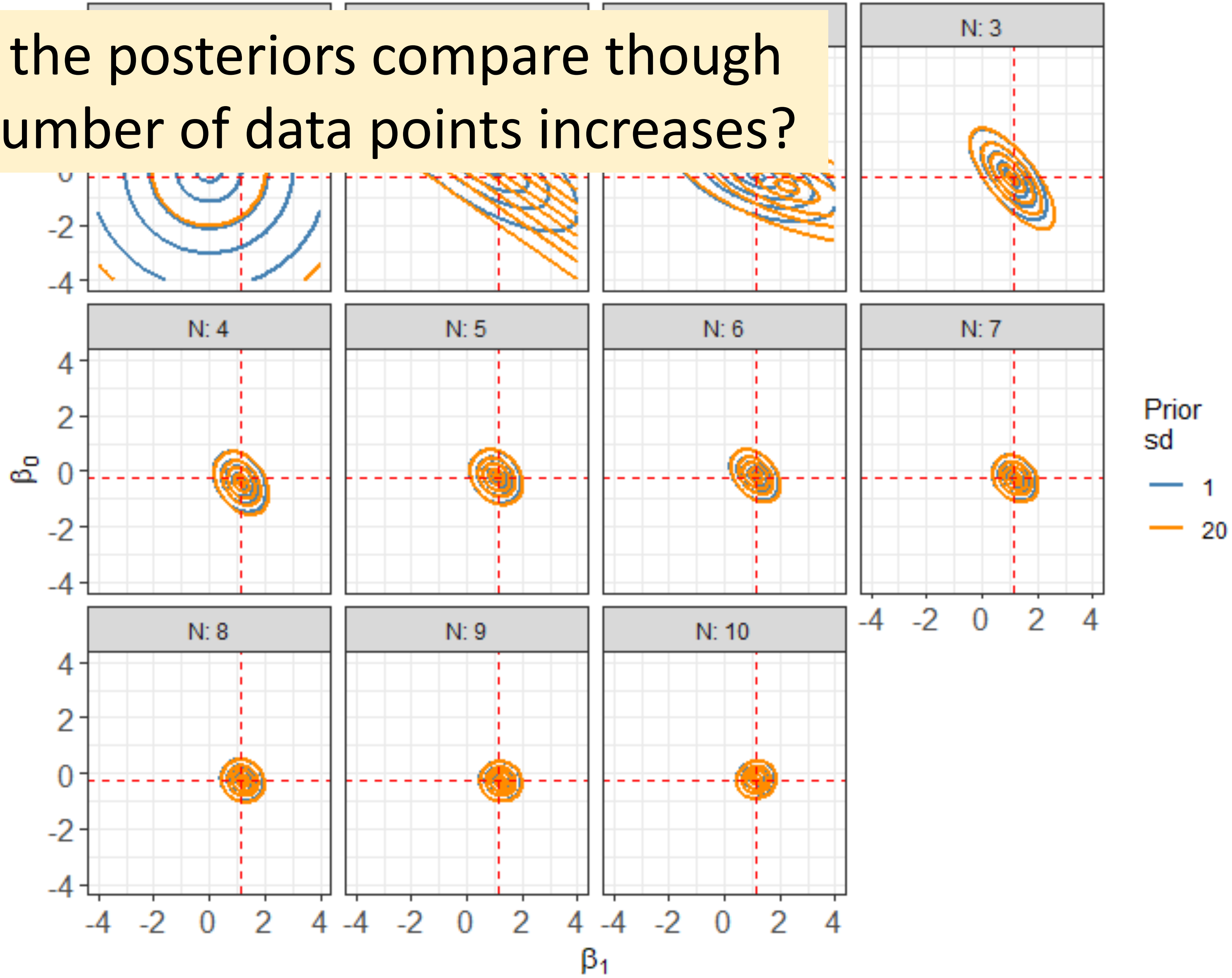
After 1 observation, the informative prior allows the trade-off to still occur.

HOWEVER, the very **high slope** and very **high intercept** values have been **RULED OUT**!

The informative prior acts to **CONSTRAIN** the parameters away from **extreme** or large values!

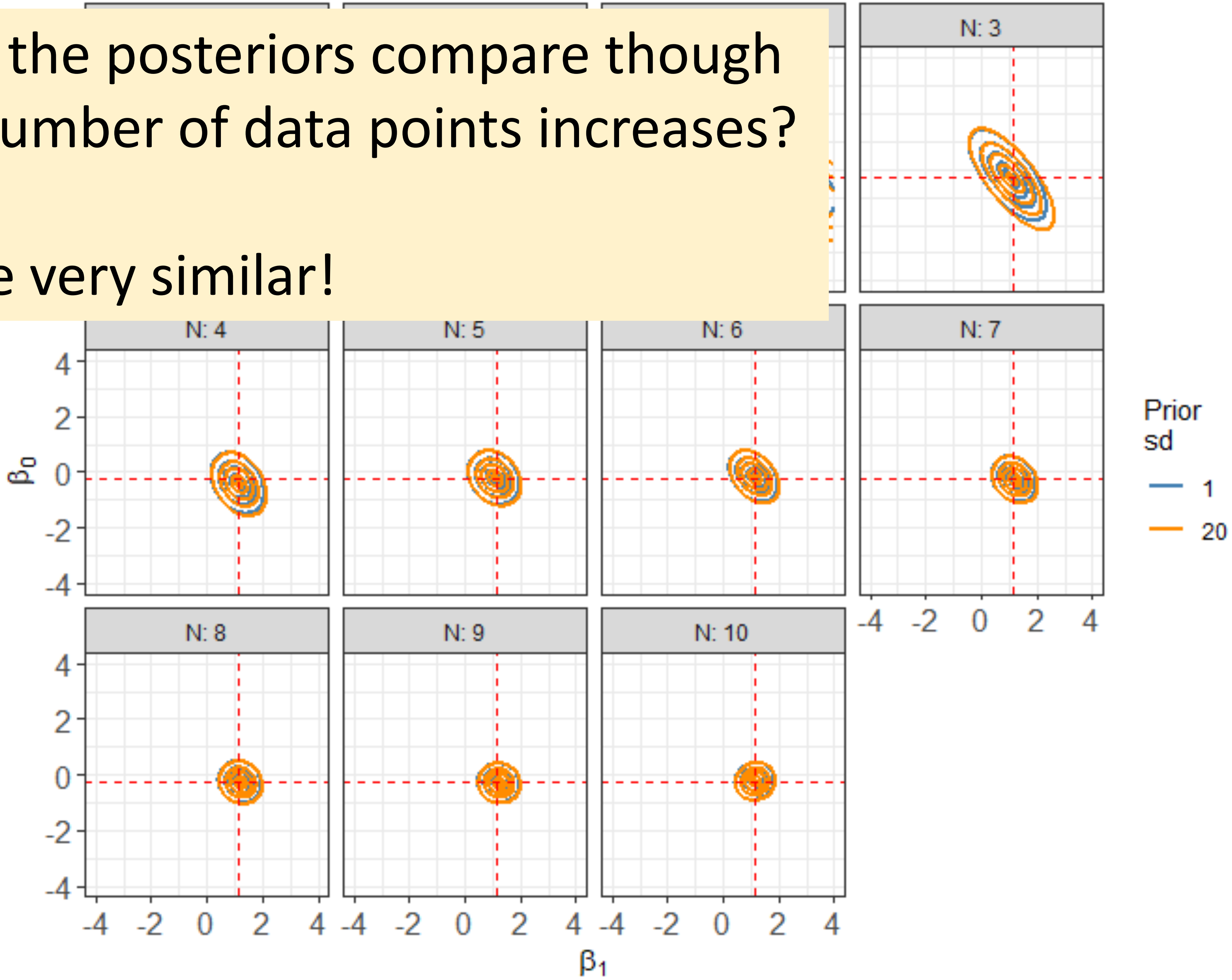
Makes it easier to “see” the mode and reduces uncertainty!!!.

How do the posteriors compare though as the number of data points increases?



How do the posteriors compare though as the number of data points increases?

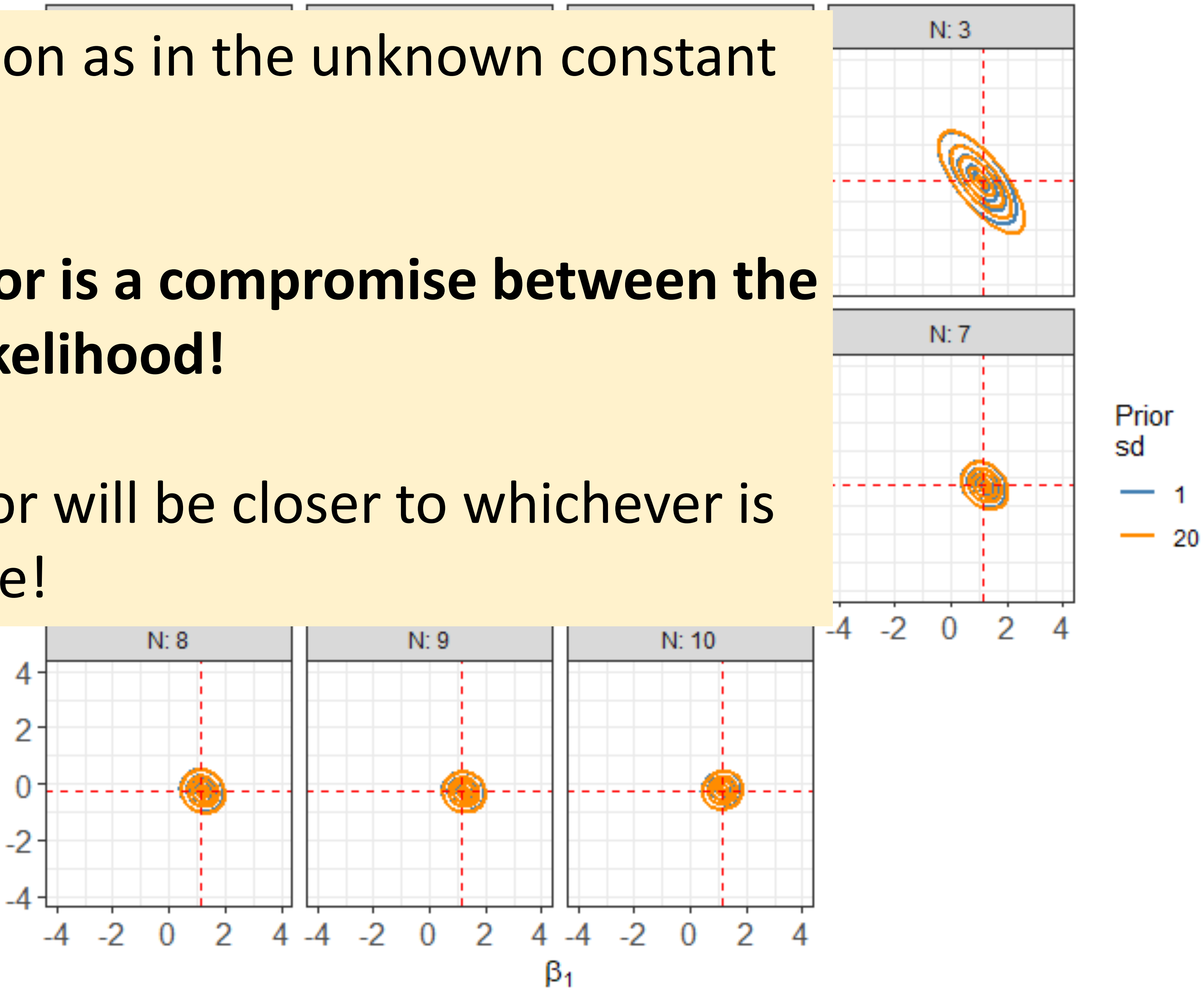
They are very similar!



Same intuition as in the unknown constant mean case.

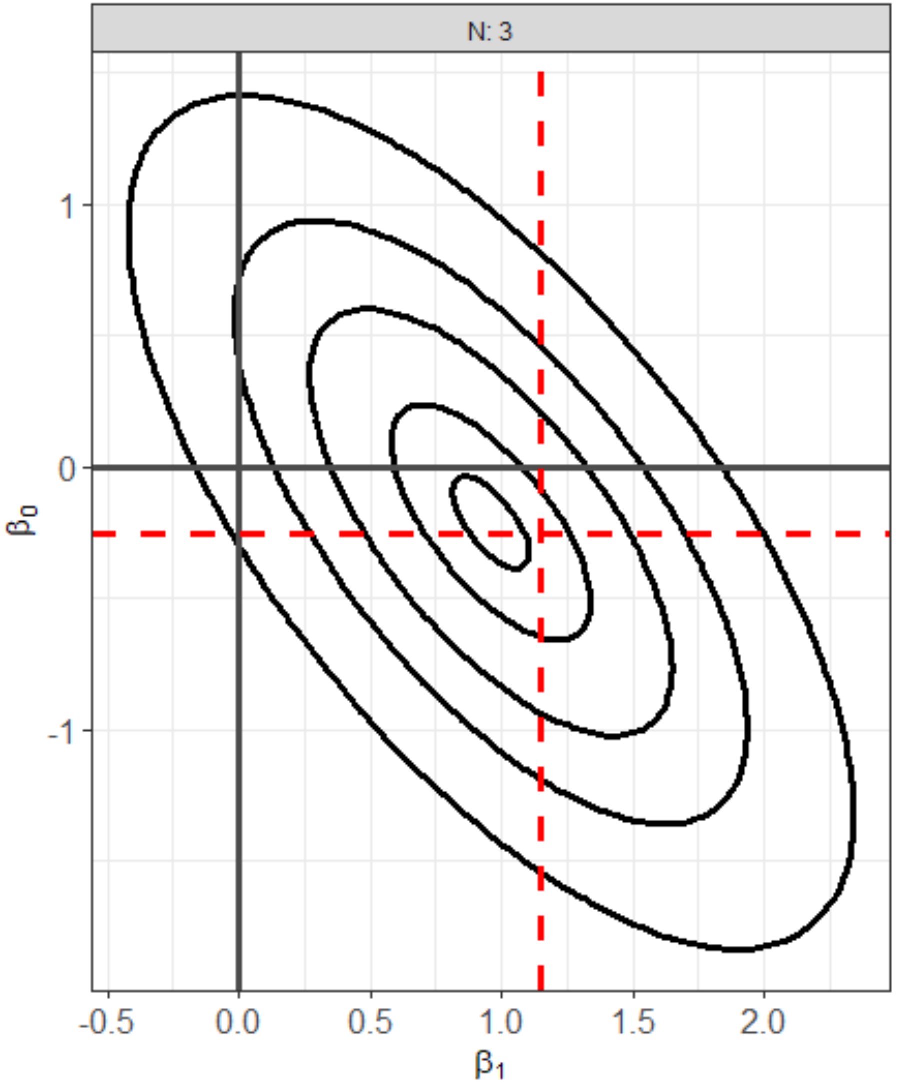
The posterior is a compromise between the prior and likelihood!

The posterior will be closer to whichever is more precise!



What's happening around a posterior mode?

Prior means denoted by horizontal and vertical grey lines.

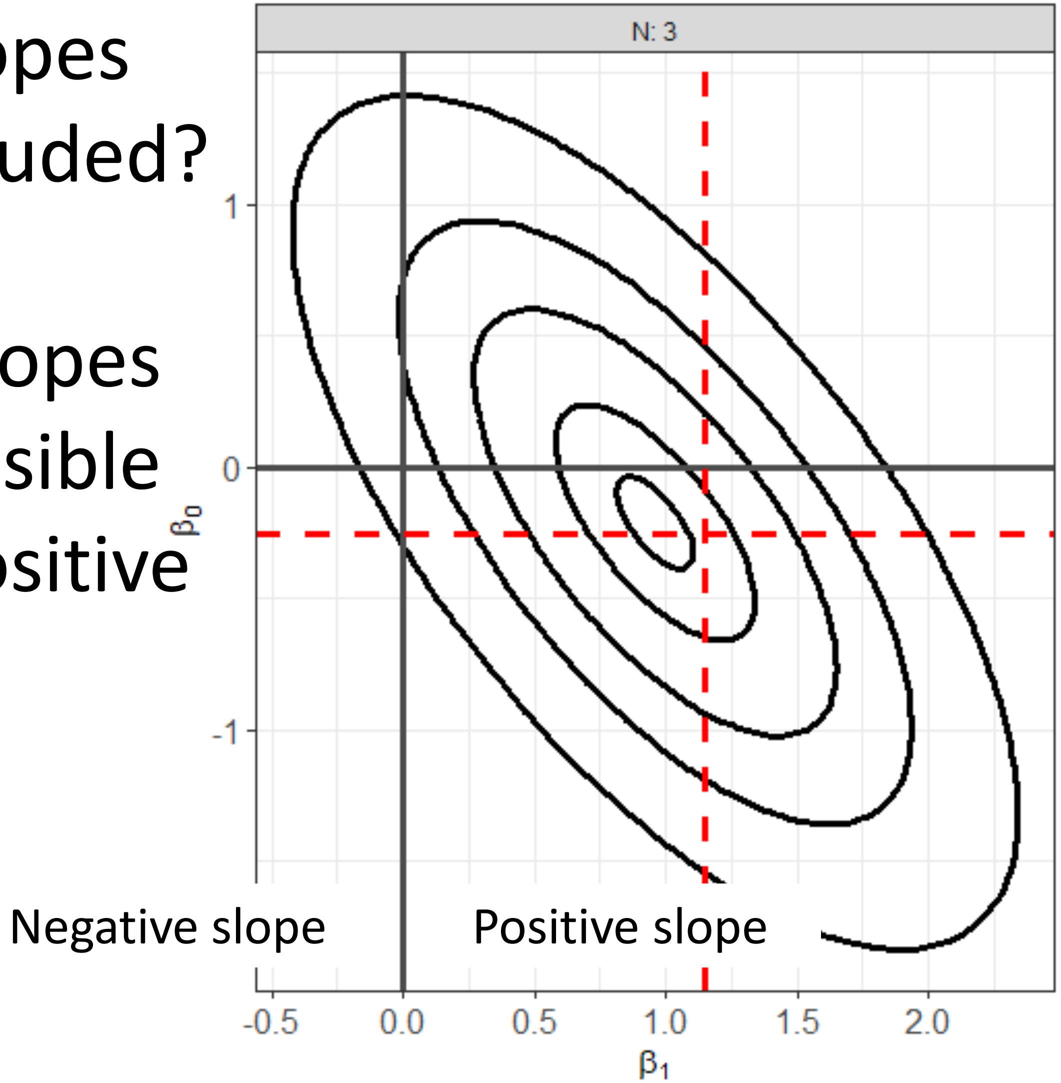


Focus on the case with the informative prior and 3 observations.

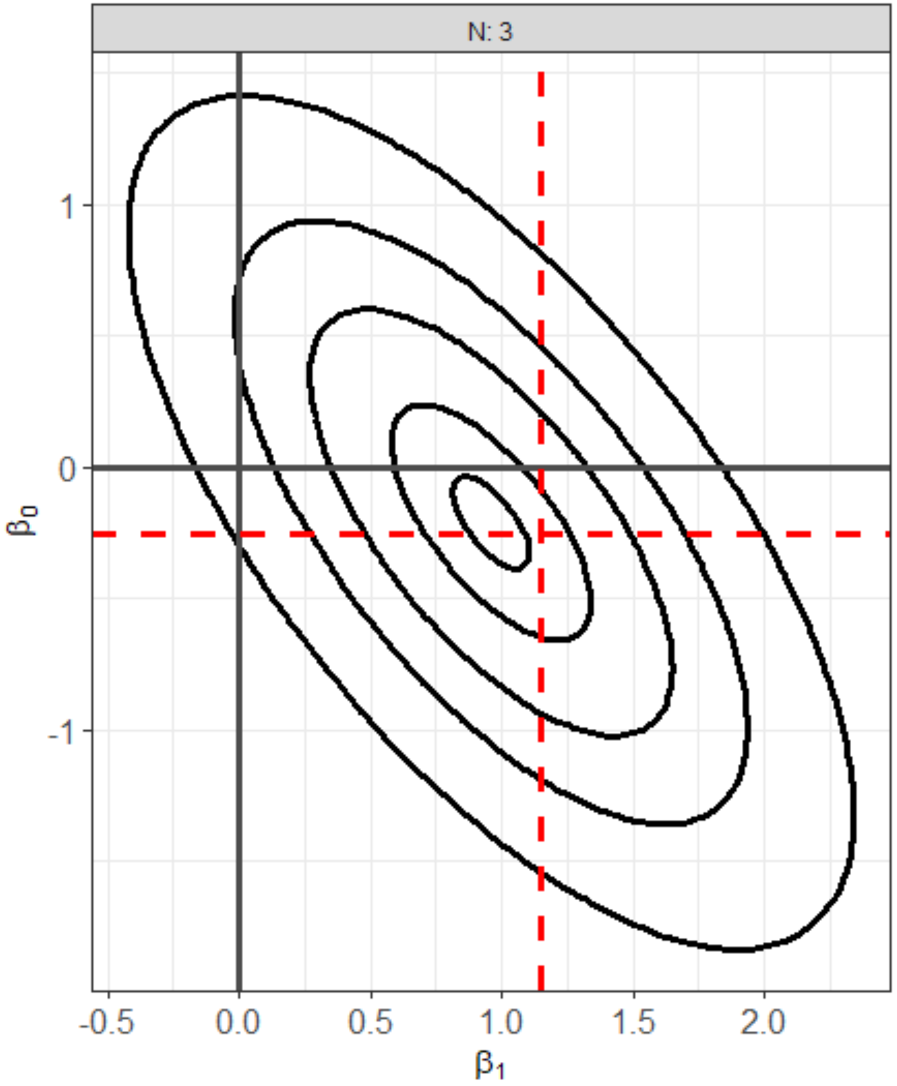
Based on the log-posterior contours…

Are negative slopes completely excluded?

NO…negative slopes are far less plausible compared to positive slopes.



Based on the log-posterior contours…



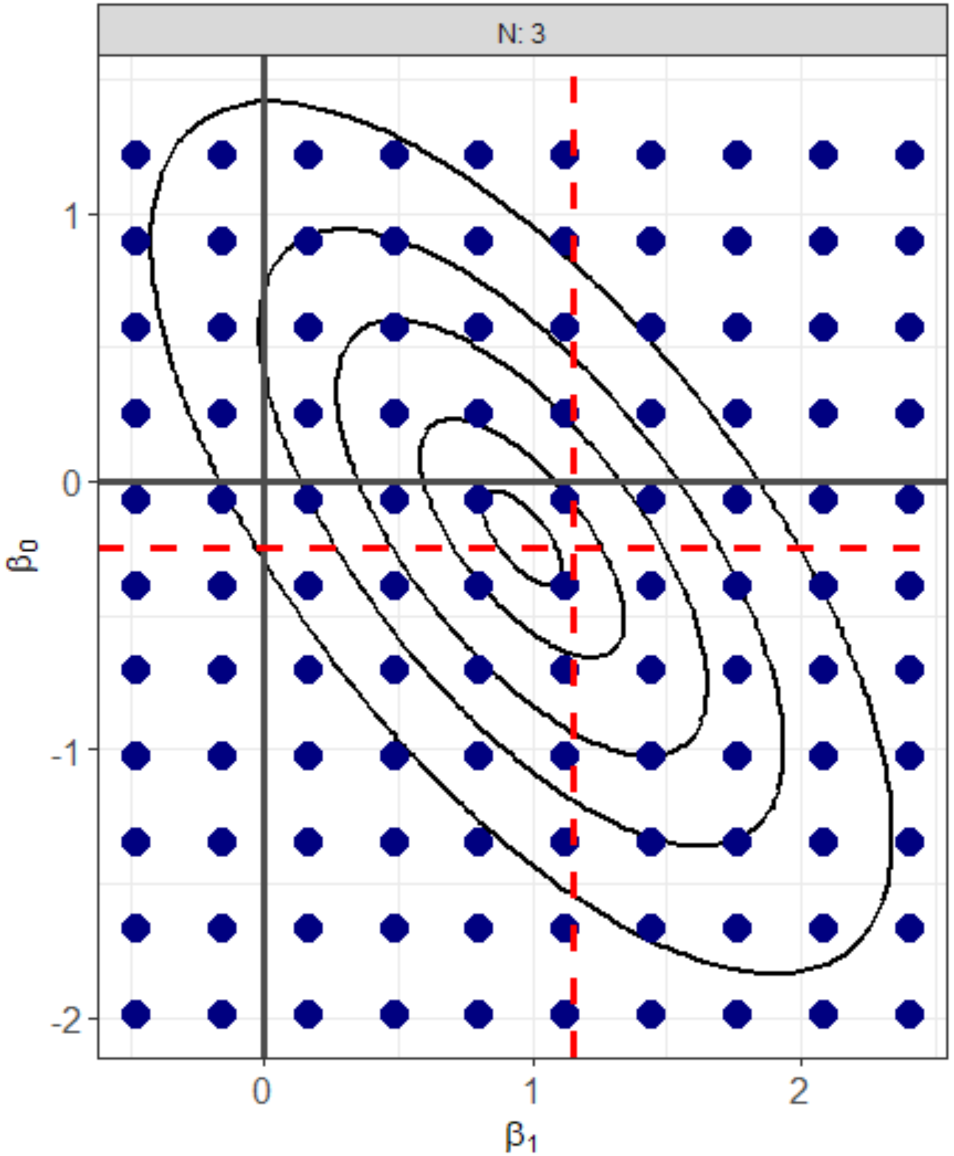
Are high intercept values and high slopes likely?

NO…high values of the intercept AND high values of the slope have been completely ruled out.

Focus on specific points in the neighborhood of the posterior

The navy blue markers correspond to specific combinations of the intercept and slope.

Each combination represents a separate line.

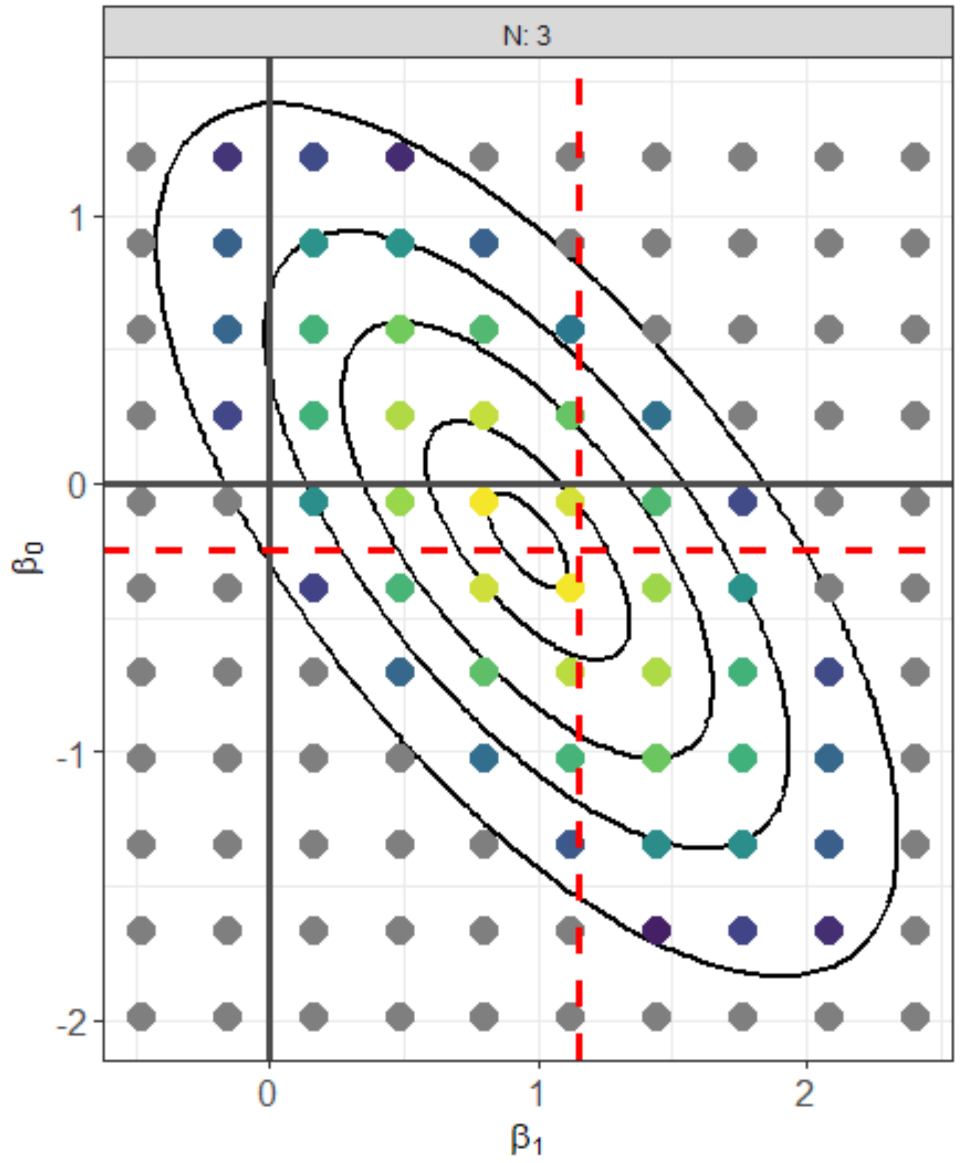


Focus on specific points in the neighborhood of the posterior

The marker color now denotes the log-posterior value of each combination.

Greyed out markers are combinations that are considered very unlikely.

The brightest yellow markers are those closest to the posterior mode.

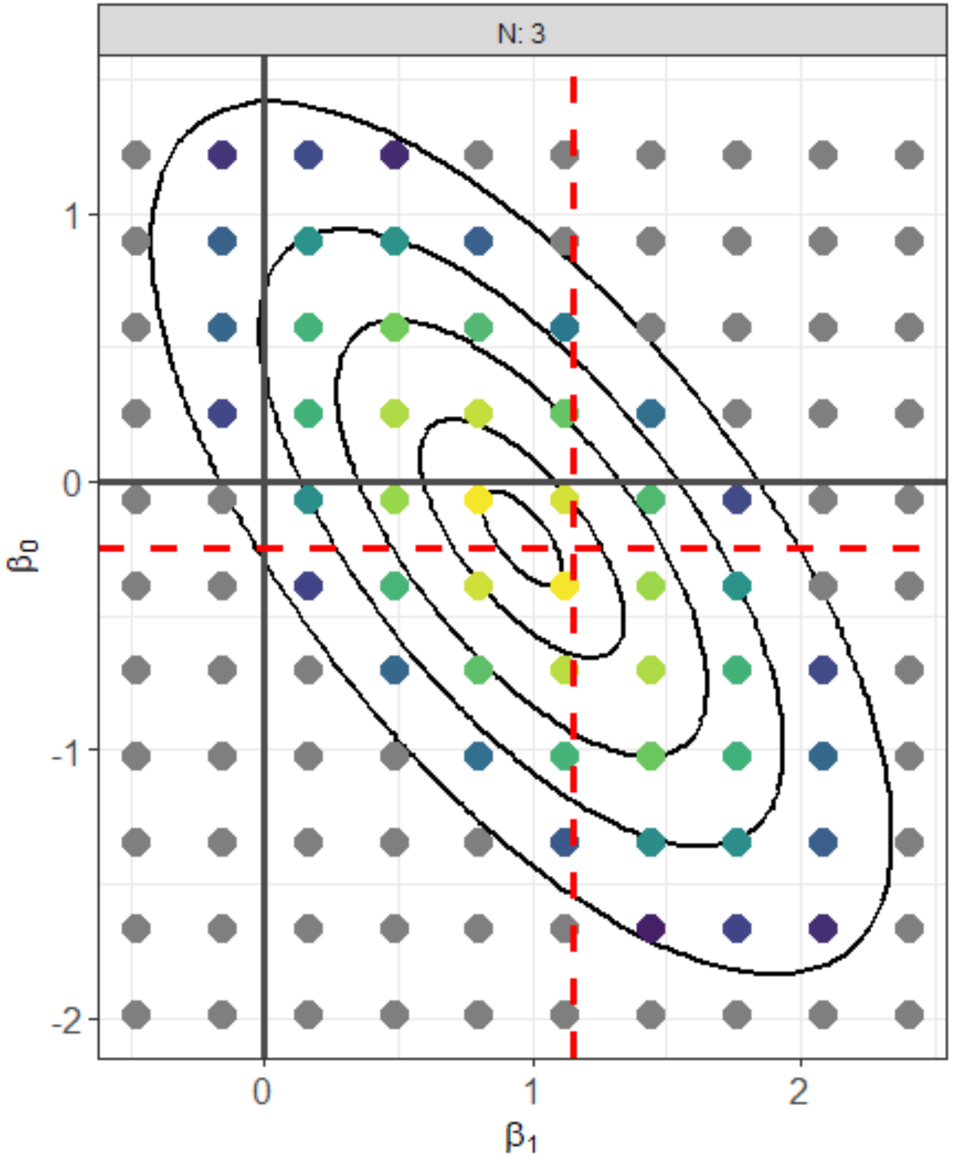


Focus on specific points in the neighborhood of the posterior

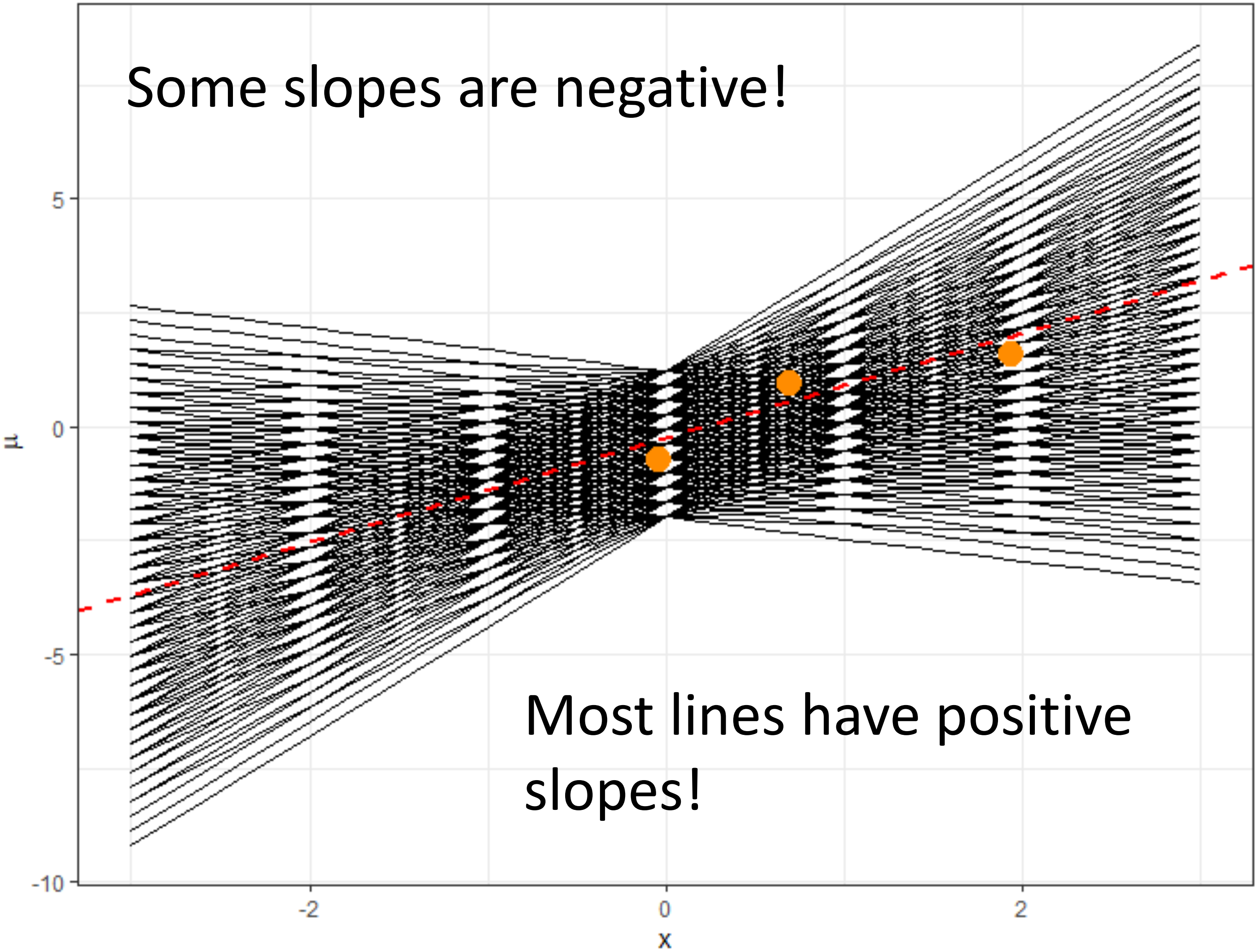
The marker color now denotes the log-posterior value of each combination.

As we saw high log-posterior values are related to LOW SSE!!!!

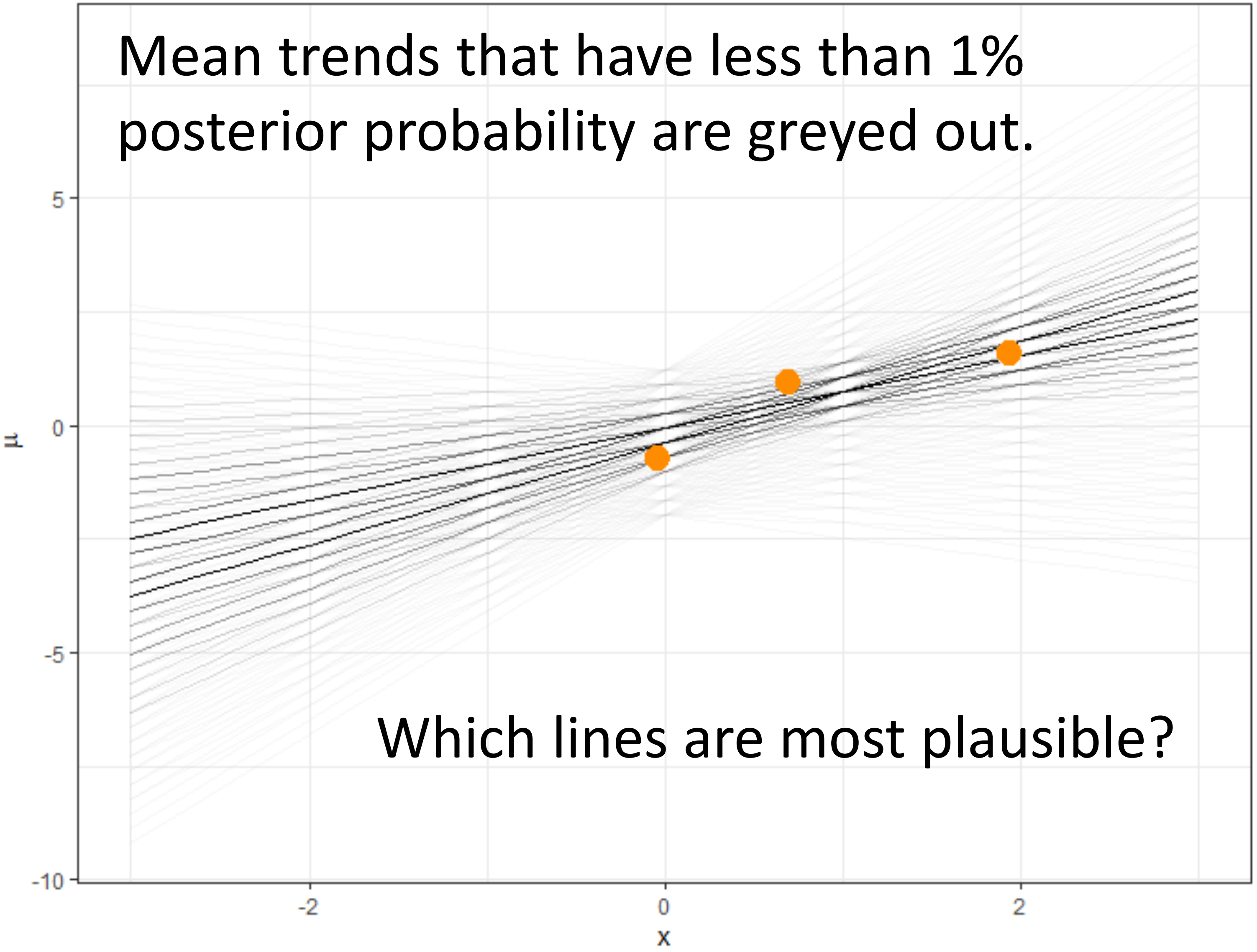
The bright yellow markers are those combinations that yield the lowest error relative to the data!



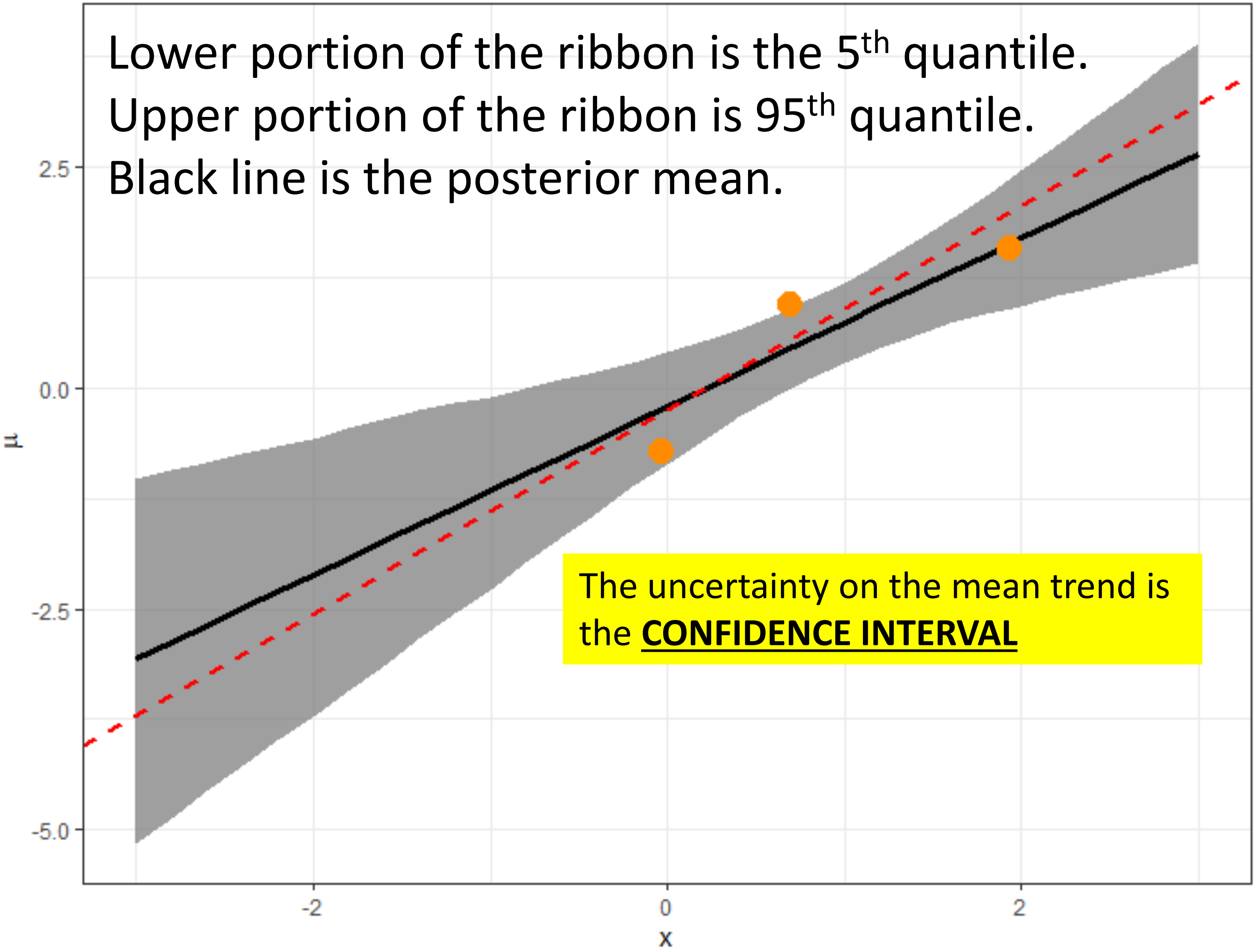
Calculate the mean trend associated with each (β_0, β_1) combination shown in the previous grid



We know which combinations are most probable because of the posterior!



We can summarize the posterior distribution of plausible mean trends



Back to the posterior distributions…what shapes do they remind you of?